Global Facility for Disaster Reduction and Recovery

A global response to reduce the risks of disasters to sustainable development

Asia Leadership Development Training in Disaster Risk Reduction and Climate Change Adaptation

Incheon, South Korea

April 26-29, 2011

DRR and Adaptation to Climate Risk and Climate Change

NATIONAL EMERGENCY MANAGEMENT AGENCY · 소방방재청 ·

GFDRR Global Facility for Disaster Reduction and Recovery

GFDRR is able to help developing countries reduce their vulnerability to natural disasters and adapt to climate change, thanks to the continued support of its partners: ACP Secretariat, Arab Academy, Australia, Bangladesh, Belgium, Brazil, Canada, Colombia, China, Denmark, Egypt, European Union, Finland, France, Germany, Haiti, India, Ireland, Italy, Japan, Luxembourg, Malawi, Mexico, The Netherlands, New Zealand, Norway, Portugal, Saudi Arabia, Senegal, Spain, South Africa, South Korea, Sweden, Switzerland, Turkey, United Kingdom, United States, Vietnam, Yemen, IFRC, UNDP, UN/International Strategy for Disaster Reduction and The World Bank.

From a development perspective, integrating CCA analysis and measures in DRR interventions is increasingly becoming a basic issue of due diligence, while CCA investments that do not simultaneously address current climate risks could fall short of meeting countries' development needs

Issues to be covered

Is the changing climate affecting disaster risk and risk management?

What can we learn from climate science and climate modelling?

What is to come?

What do these projections mean for DRM?

Limits to what we can know and the downscaling issue

Important similarities and differences between DRR & CCA

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Is the changing climate affecting disaster risk and risk management?

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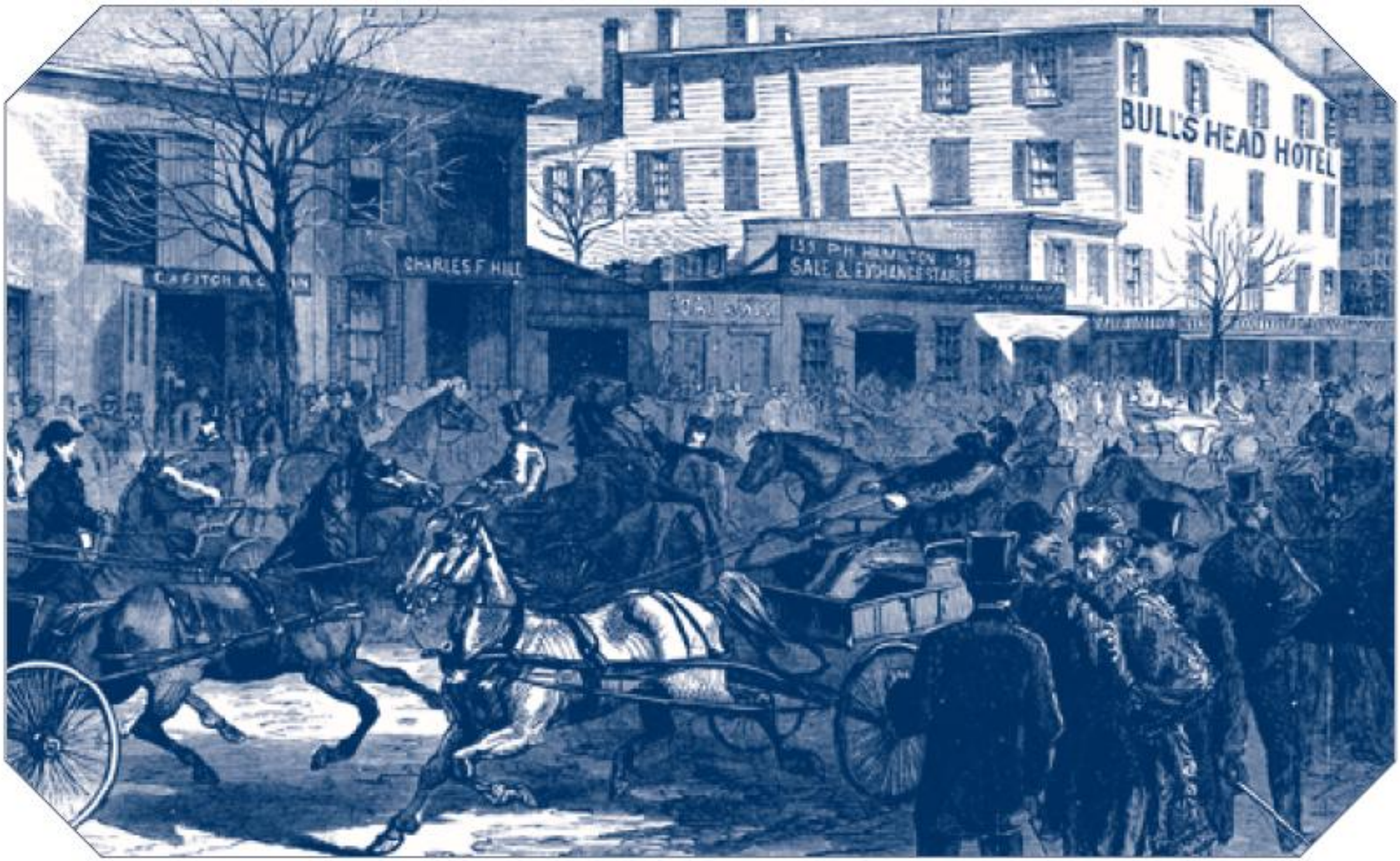
Important similarities and differences between DRR & CCA

What was the dominant topic of the First World Planning Conference in NY in 1898?

- Housing?
- Landuse?
- Infrastructure?

Immigration?
Crime?
Something else?



Approximately 200,000 horses deposited 2000 tonnes of manure each day on the city streets; plus 50 or more horses died in the streets

In 1894, the *Times of London estimated* that by 1950 every street in the city would be buried nine feet deep in horse manure.

"A public health and sanitation crisis of almost unimaginable dimensions looms"

Number of horses was increasing at a rate far higher than the population as city populations became wealthier and demanded more products.

Each horse required 2 ha of land – enough to sustain about 6 people

A cautionary message …

BBC Summary - "…according to scientists, up until now, weather-related natural hazards, with their inherent variability, may have not been increasing – it's the vulnerability of the population that has seen a sharp rise."

Study Finds No Link Tying Disaster Losses to Human-Driven Warming
By ANDREW C. REVKIN
NYT

NPR -The paper "… finds no causal link between increases in disaster losses and man-made climate change."

AMERICAN
METEOROLOGICAL
SOCIETY

Bulletin of the American Meteorological Society

Bulletin of the American Meteorological Society 2010
doi: 10.1175/2010BAMS3092.1

Have disaster losses increased due to anthropogenic climate change?

Laurens M. Bouwer
Institute for Environmental Studies, Vrije Universiteit, Amsterdam, The Netherlands

Conclusions

- The analysis of twenty-two disaster loss studies shows that economic losses from various weather related natural hazards, such as storms, tropical cyclones, floods, and small-scale weather events such as wildfires and hailstorms, have increased around the globe.

- The studies show no trends in losses, corrected for changes (increases) in population and capital at risk, that could be attributed to anthropogenic climate change.

- Therefore it can be concluded that anthropogenic climate change **so far** has not had a significant impact on losses from natural disasters.

- Considerable uncertainties remain in some of these studies, as exposure and vulnerability that influence risk can only be roughly accounted for over time.

- In particular the potential effects of past risk reduction efforts on the loss increase are often ignored, because data that can be used to correct for these effects is not available.

Conclusions

- The analysis of twenty-two disaster loss studies shows that
  … l hazards,
  … cale weather
  … ed around the
  … hanges
  … d be
  … mate change
  … natural
  disasters.

- Analysed losses only
- Removed what are known to be the biggest drivers of losses (capital and people at risk)
- Then found no significant remaining trend
- But failure to detect a significant statistical trend does not show the non-existence of a trend

"The bottom line? Regardless of what happens due to global warming, on a crowding, urbanizing planet, increased exposure to, and losses from, nature's hard knocks are a sure thing if people keep settling in harm's way."
Andrew Revkin

- Considerable uncertainties remain in some of these studies, as exposure and vulnerability that influence risk can only be roughly accounted for over time.
- In particular the potential effects of past risk reduction efforts on the loss increase are often ignored, because data that can be used to correct for these effects is not available.

Message 1:
Be careful of the media's interpretation of the science

What have we observed?

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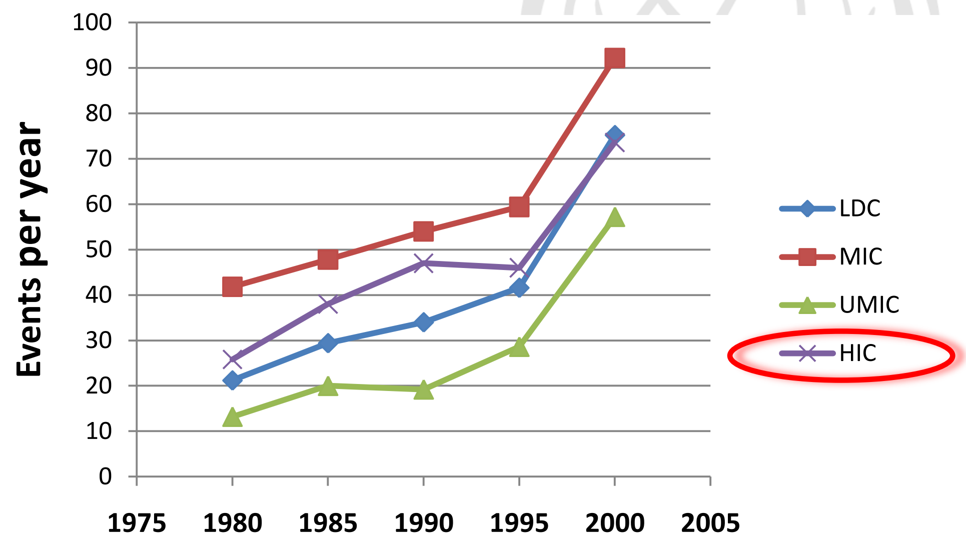
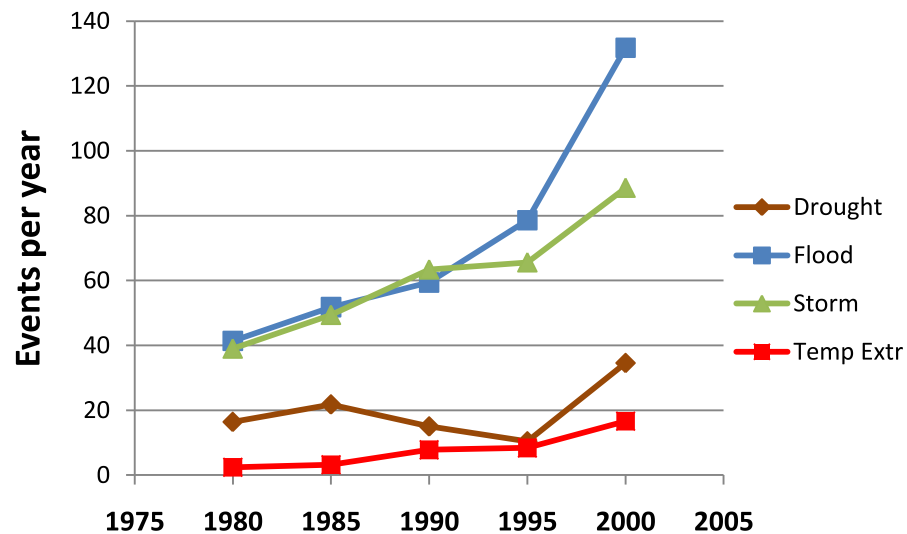
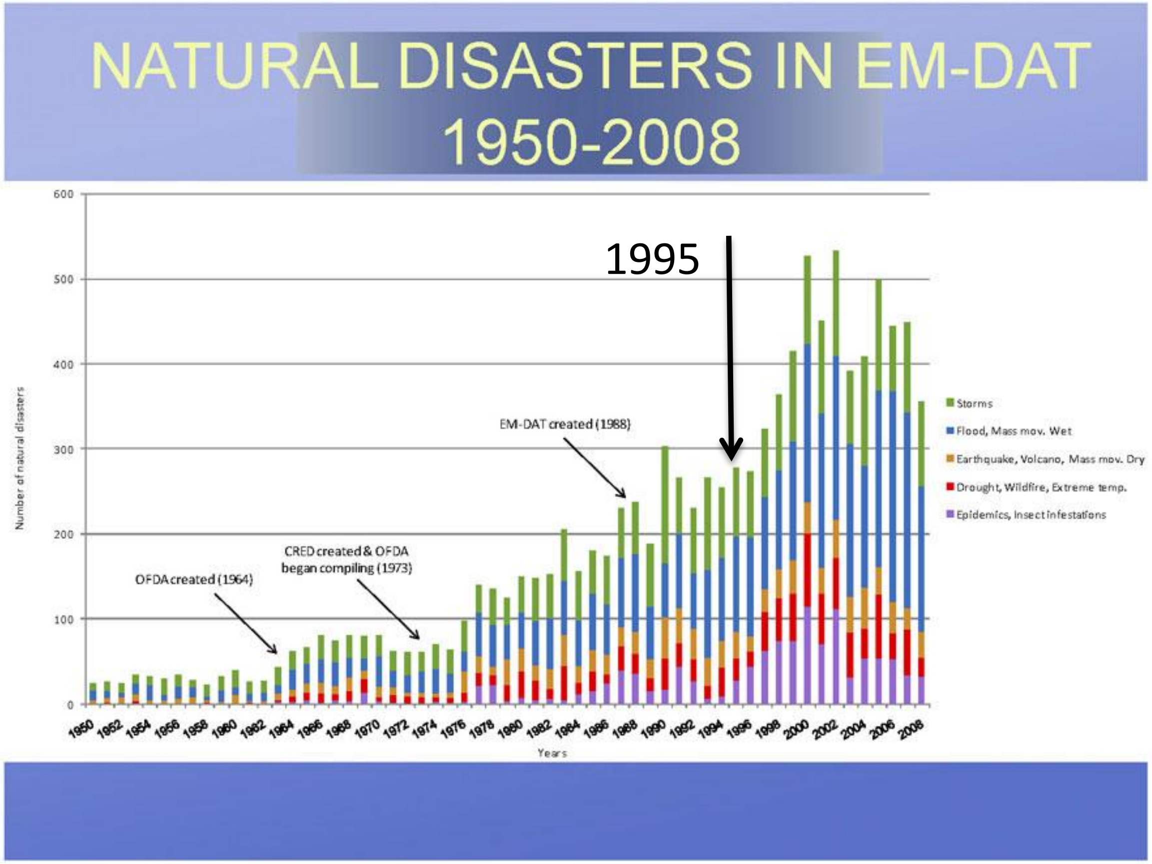
More disaster events are being reported every decade





But is this reporting or climate change – or both?



We believe that the increase seen in the graph until about 1995 is explained

- partly by better reporting of disasters in general,
- partly due to active data collection efforts by CRED and
- partly due to real increases in certain types of disasters.

We estimate that the data in the most recent decade present the least bias and reflect a real change in numbers. This is especially true for floods and cyclones. Whether this is due to climate change or not, we are unable to say.

CRED (Centre for Research on the Epidemiology of Disasters)

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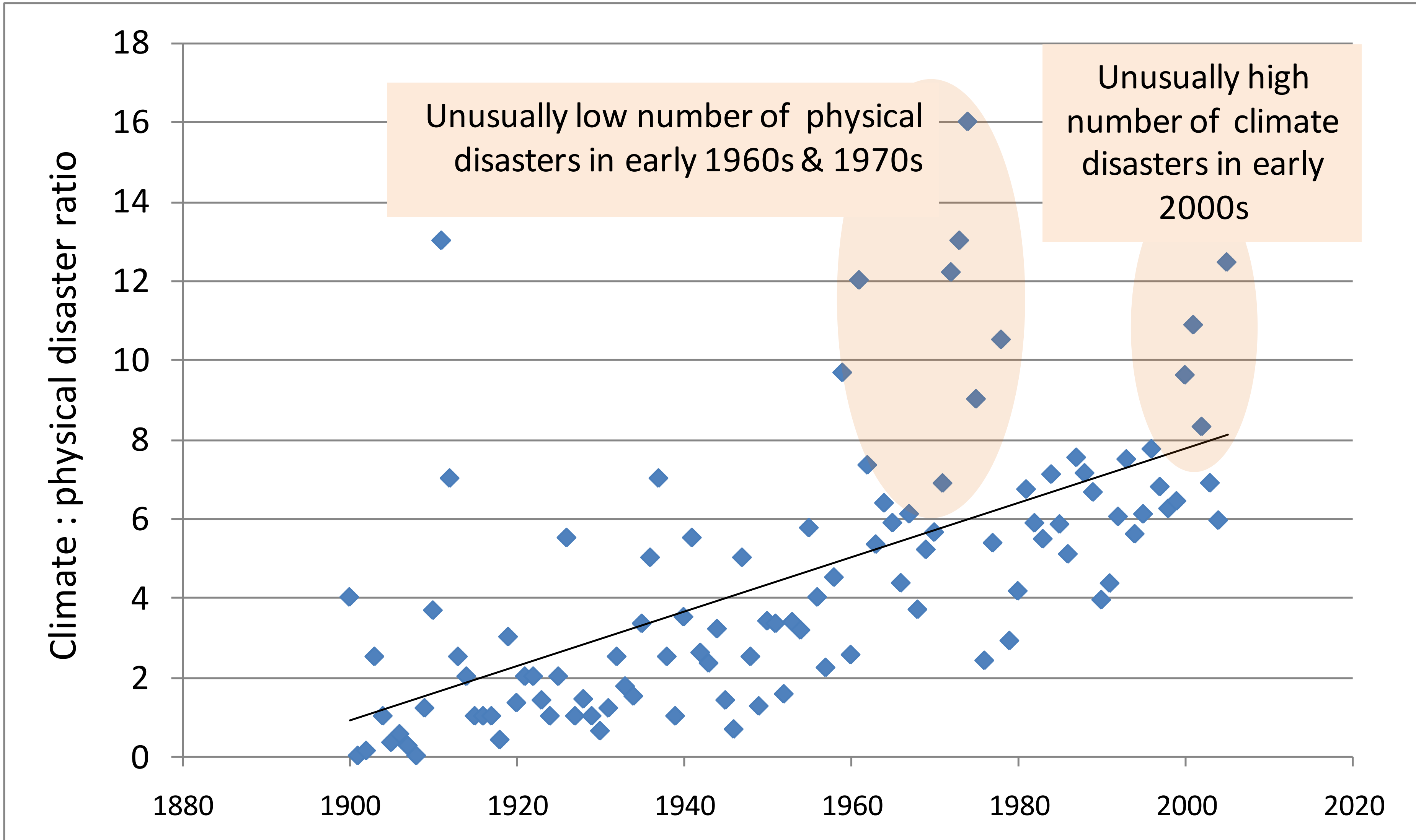
Climate related disasters are increasing faster than non-climate disasters



Message 2:
More climate related disasters are being reported every decade –
unlikely to arise only from better reporting

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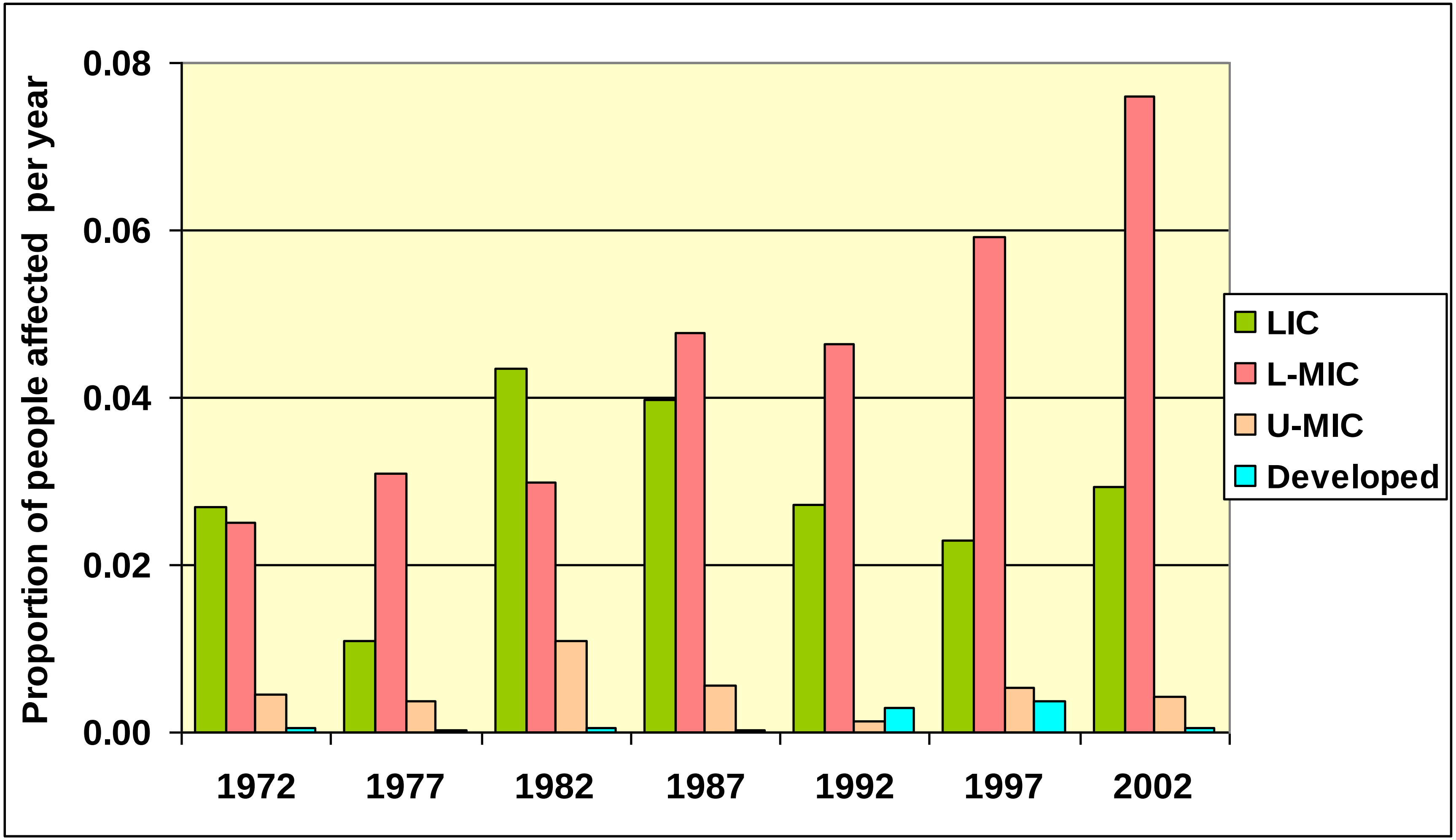
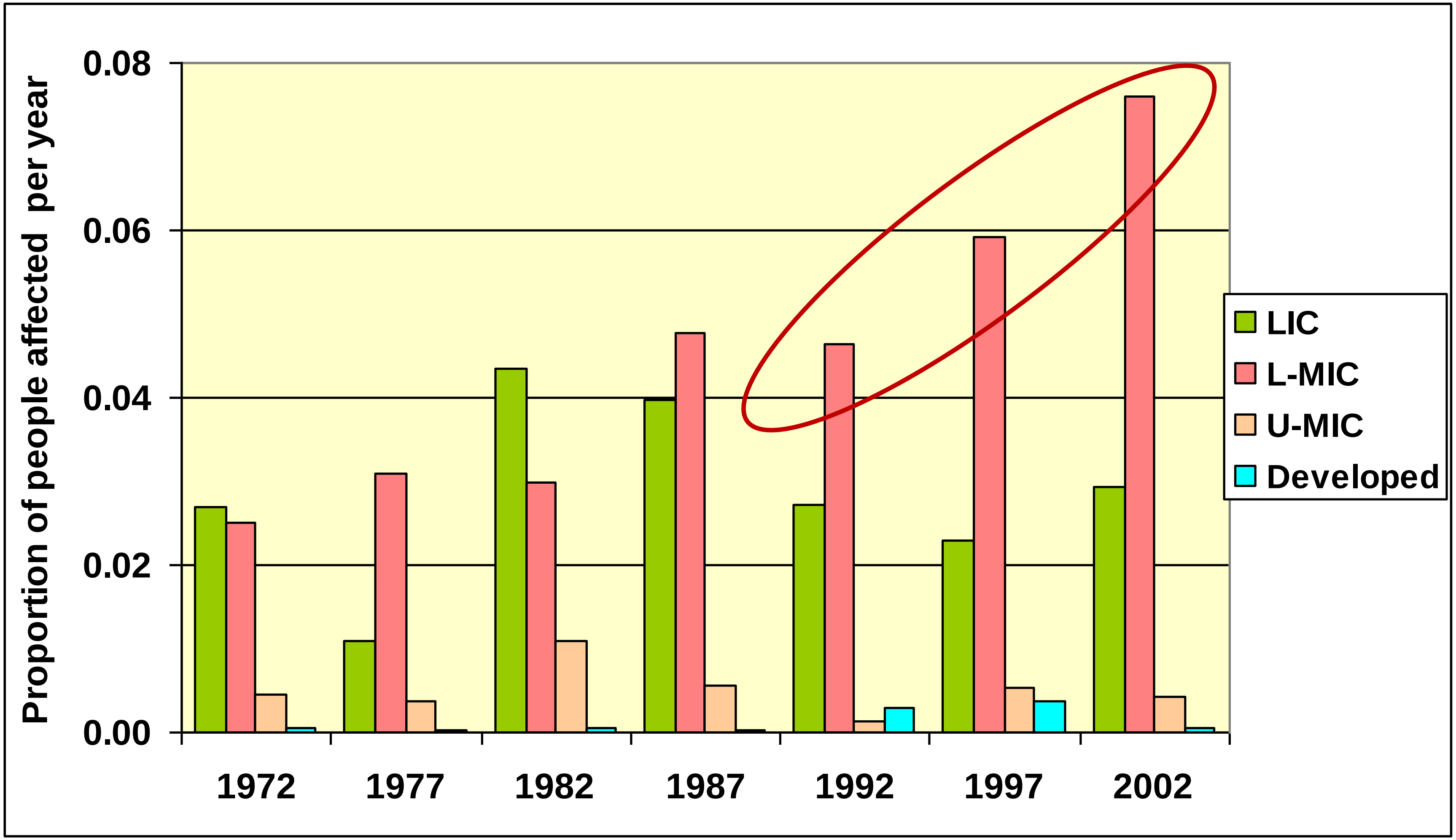
More and more people are being affected by climate related disasters each decade

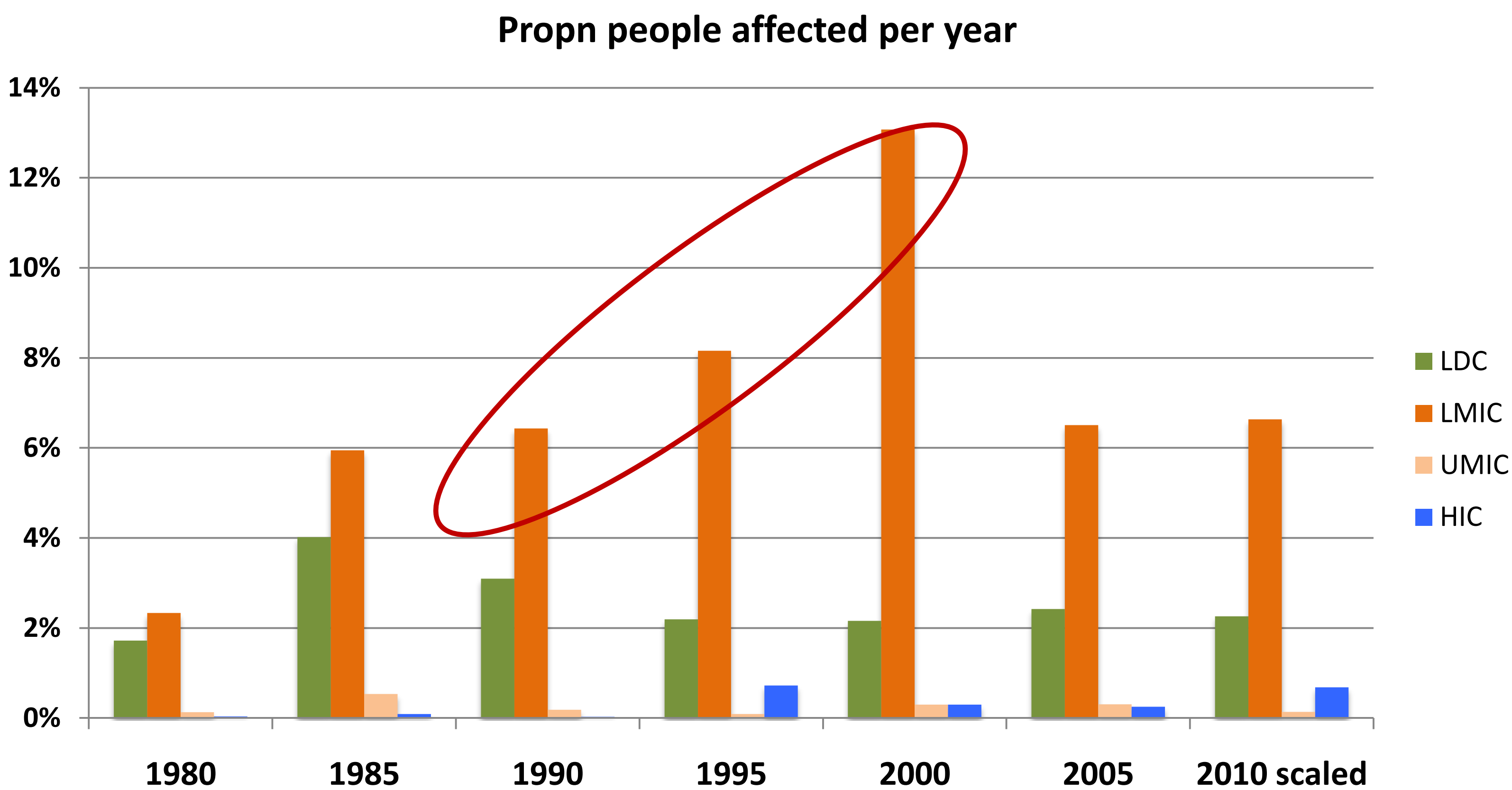


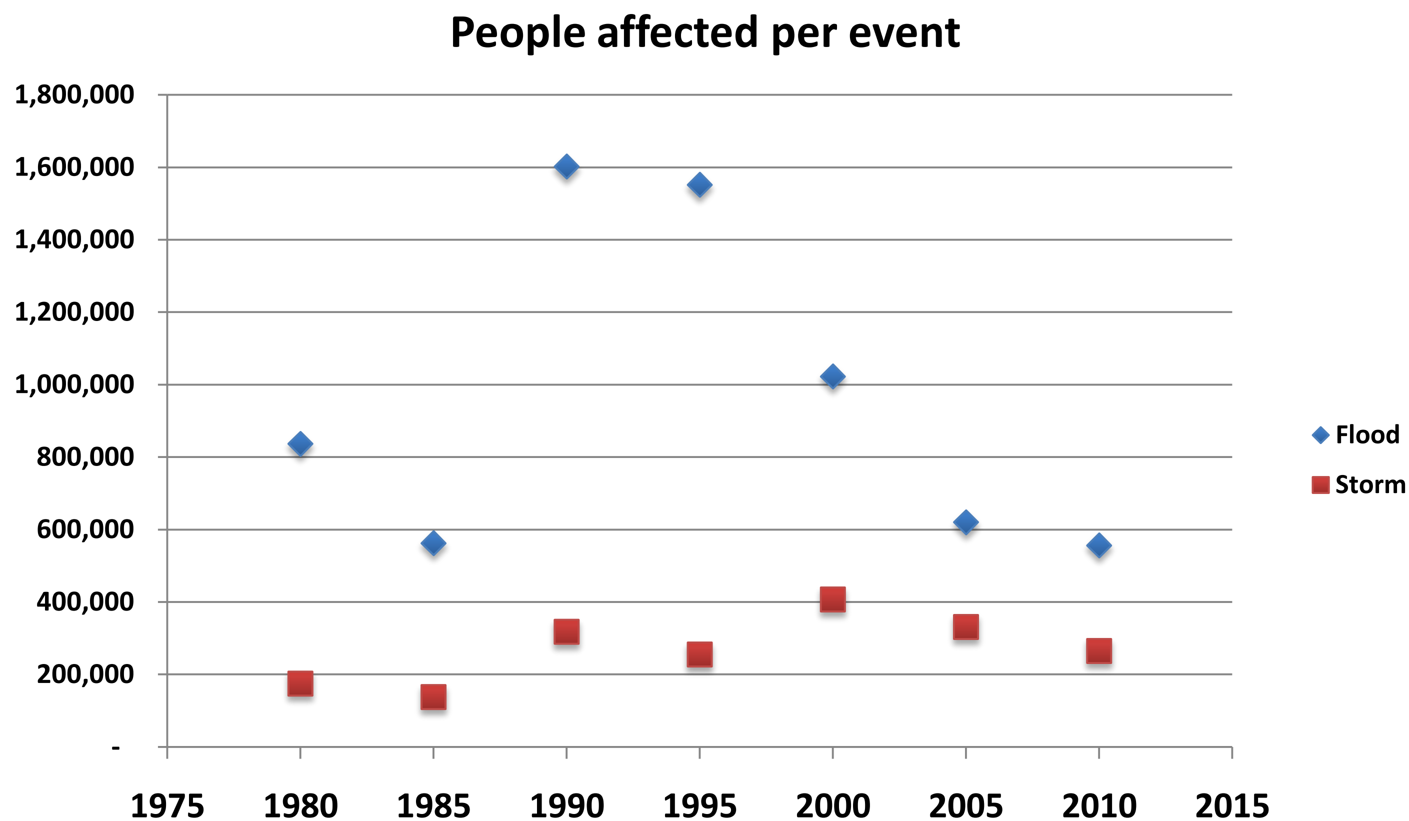
Lower Middle-Income Countries appear to be the most affected



But sometimes more data simply confuses the story!

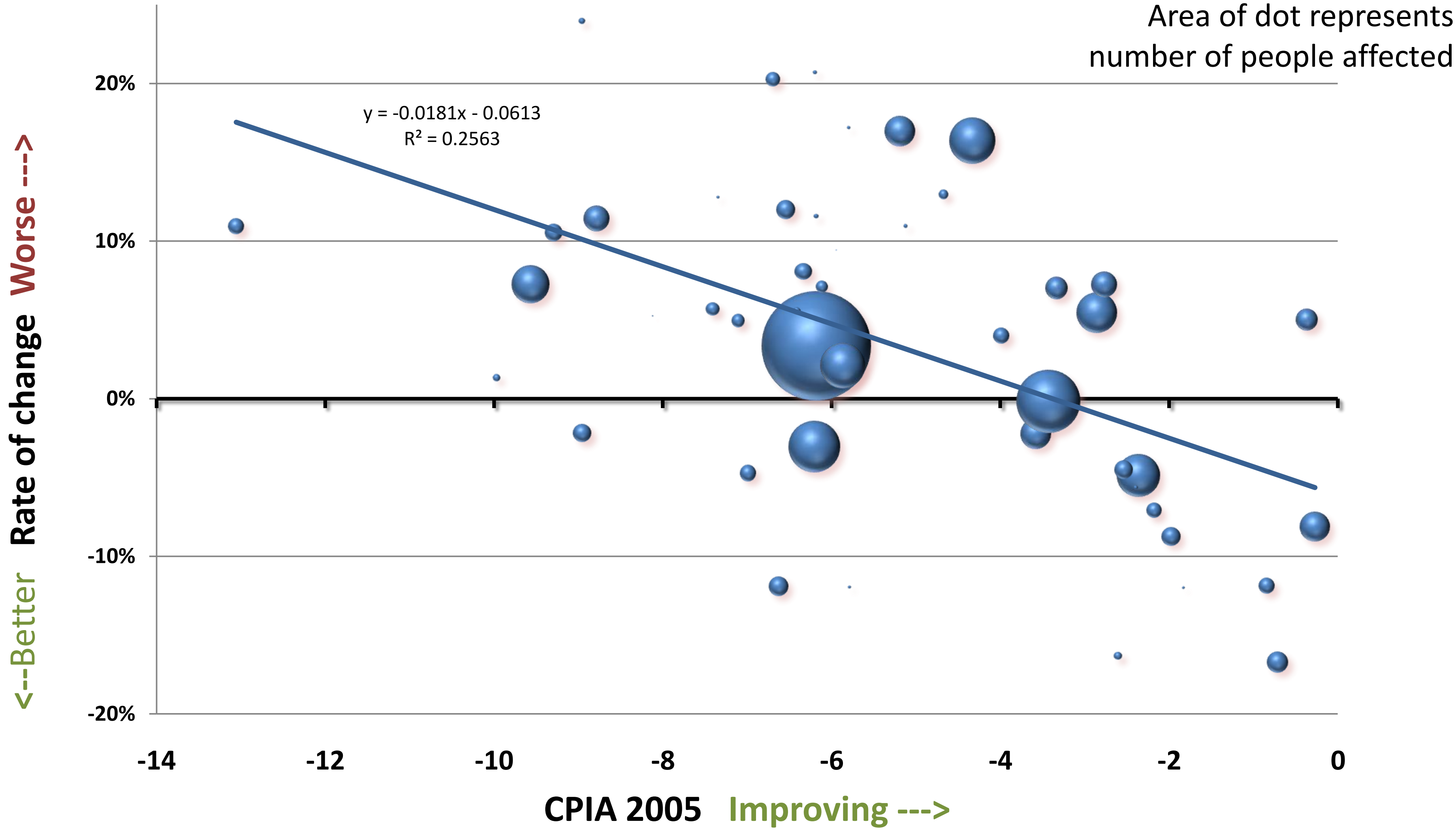
Longer data series and cleaning of CRED database




People affected per event
1,800,000
1,600,000
1,400,000
1,200,000
1,000,000
800,000
600,000
400,000
200,000
-
1975
1980
1985
1990
1995
2000
2005
2010
2015
Flood
Storm

LDCs – governance seems to matter

Percent change in people affected per 5 years



Message 3:
More people are being affected by disasters, but the impact varies with income levels

(And the data is very noisy)

What can we learn from climate science and climate modelling?

Issues to be covered

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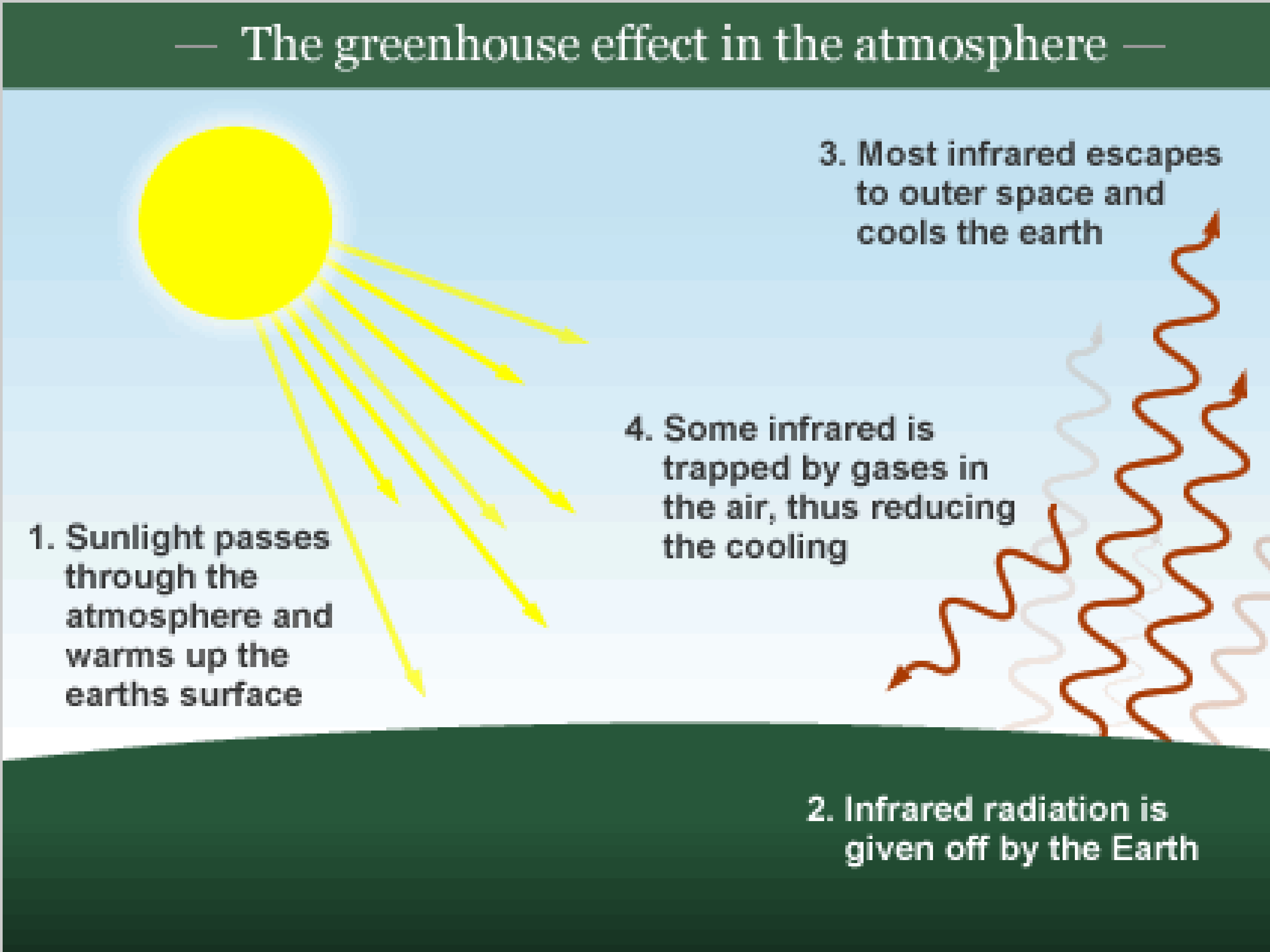
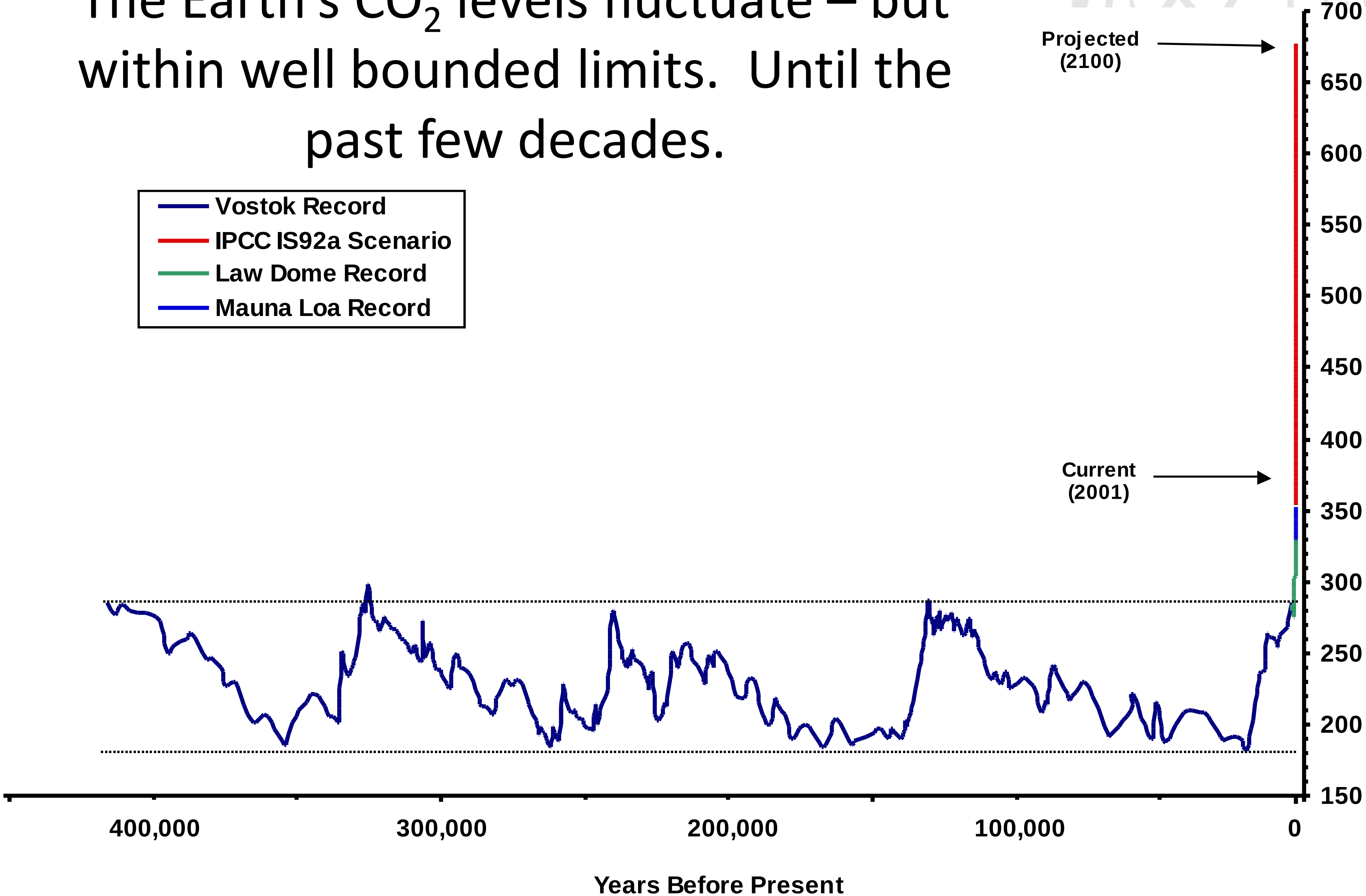
Limits to what we can know and the downscaling issue

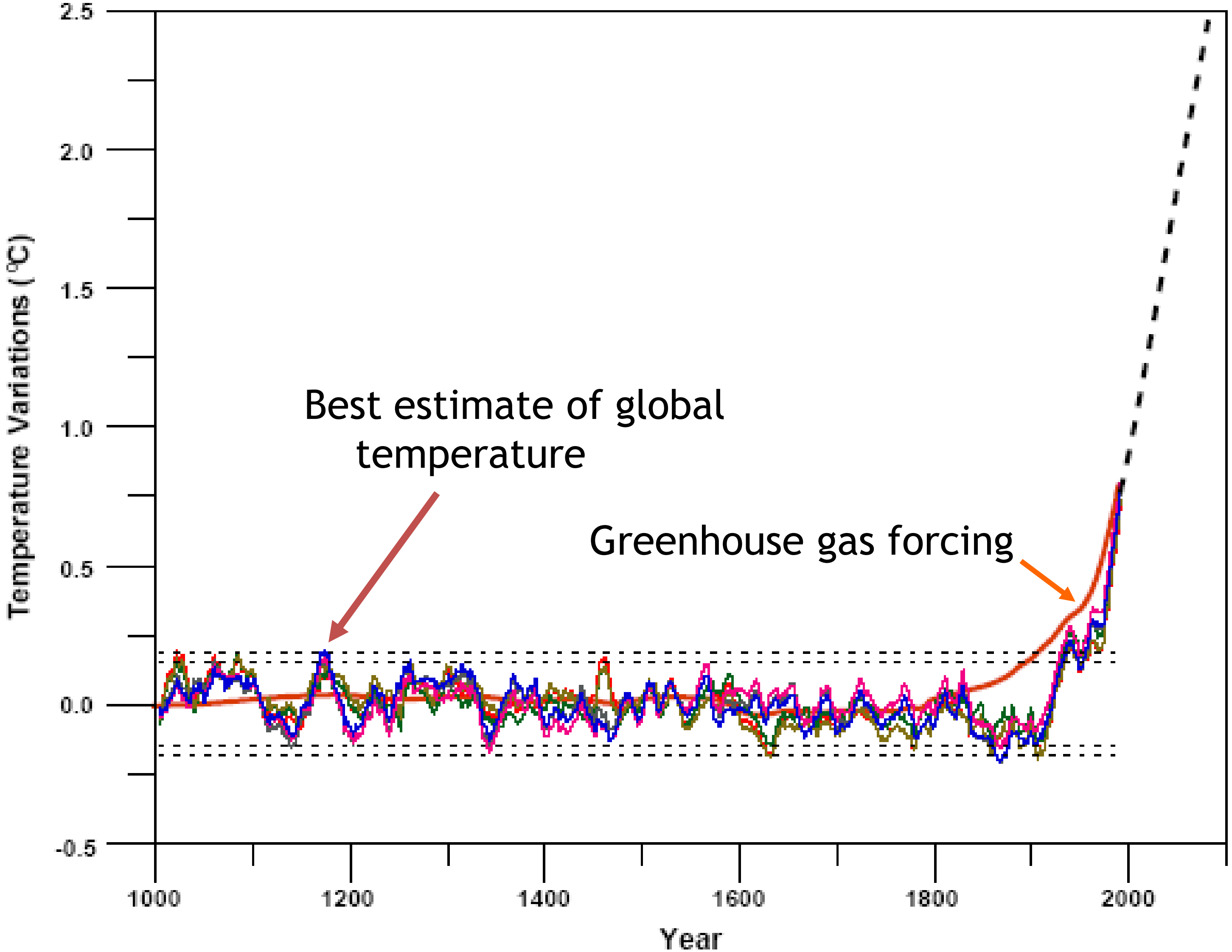
Important similarities and differences between DRR & CCA


— The greenhouse effect in the atmosphere —
1. Sunlight passes through the atmosphere and warms up the earths surface
2. Infrared radiation is given off by the Earth
3. Most infrared escapes to outer space and cools the earth
4. Some infrared is trapped by gases in the air, thus reducing the cooling

The Earth's CO_2 levels fluctuate – but within well bounded limits. Until the past few decades.

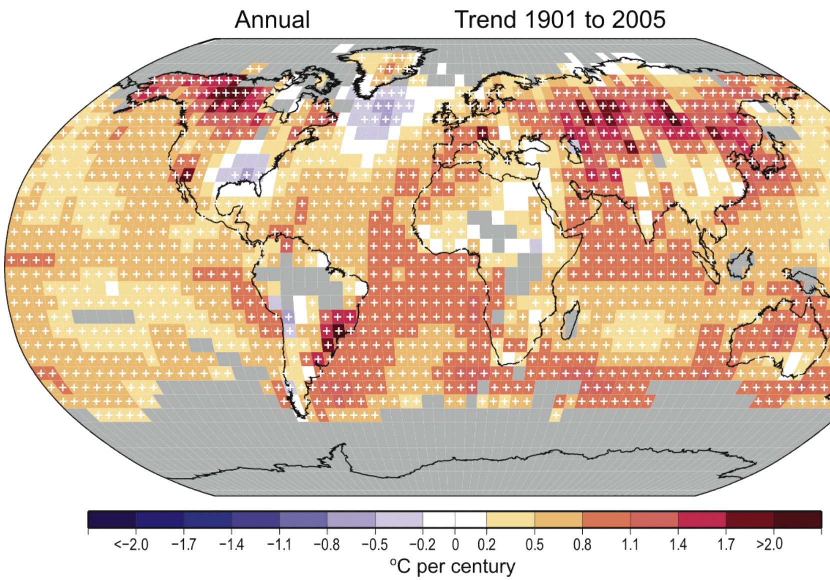


Temperatures have changed

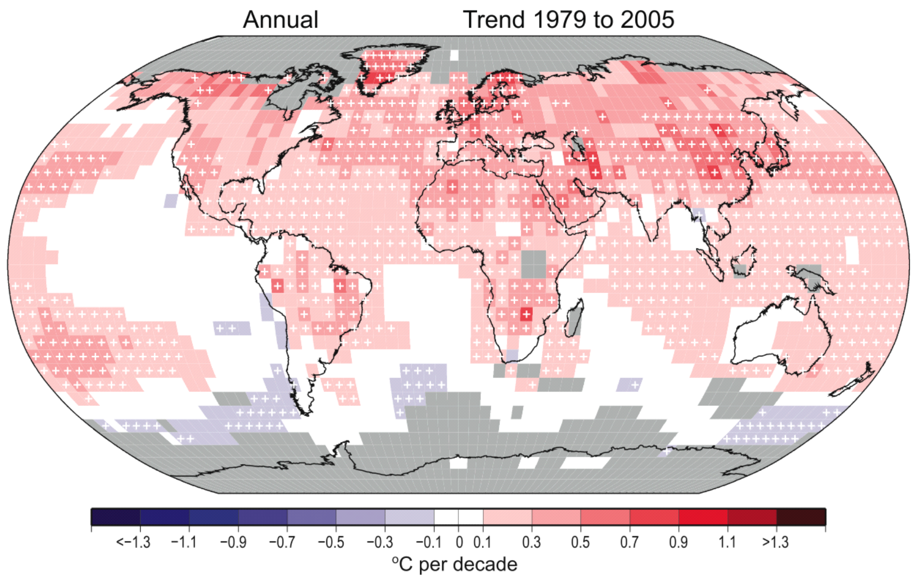


Crowley Science 2000 NOAA

Observed changes in temperature

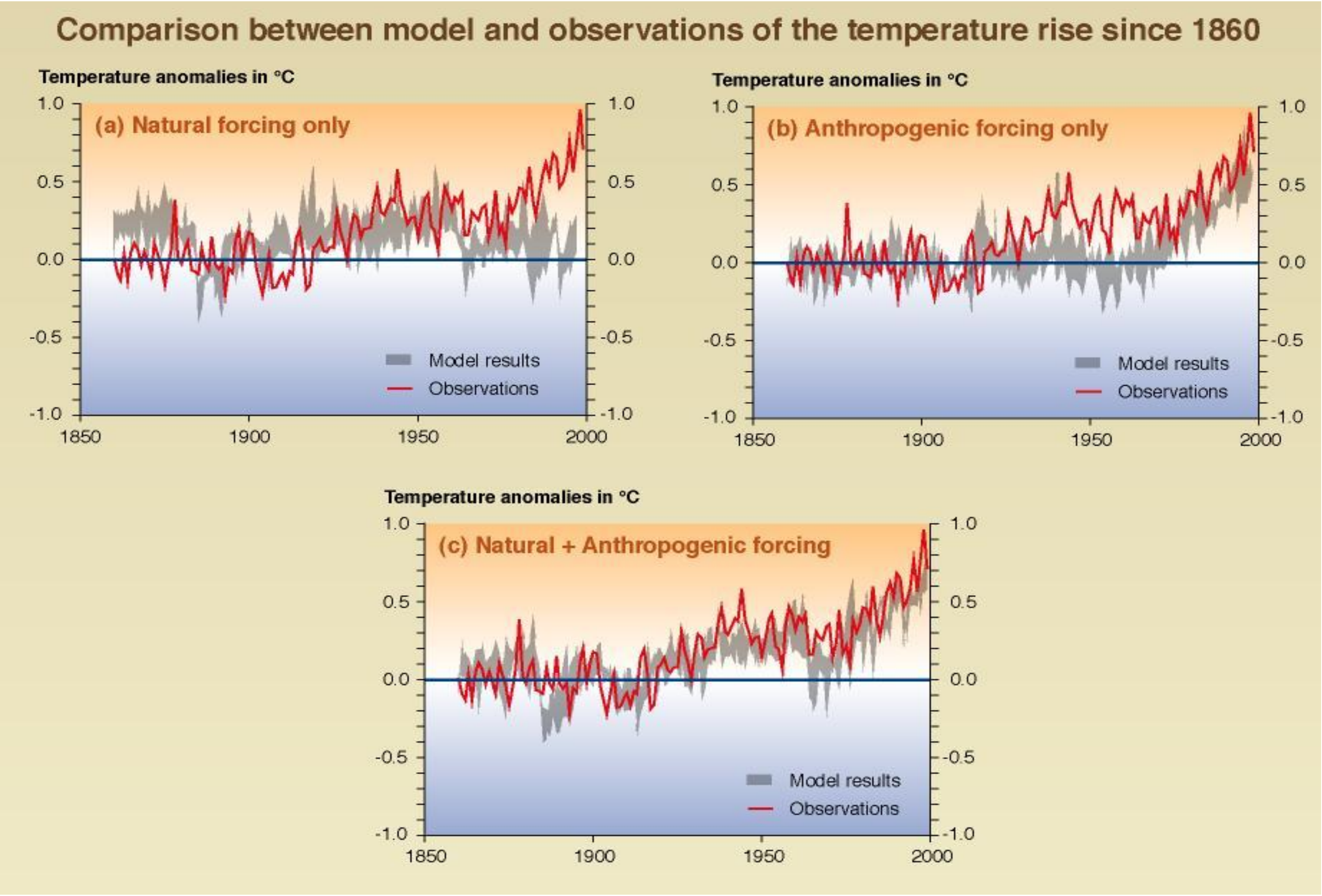


An observed temperature increase over almost the entire globe



A rapid increase in many regions in the past few decades

Temperature changes in the past few decades can be explained only by natural cycles AND human activity



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The IPCC Sequence of Key Findings……

IPCC (1990) Broad overview of climate change science, discussion of uncertainties and evidence for warming.

IPCC (1995) "The balance of evidence suggests a discernible human influence on global climate."

IPCC (2001) "Most of the warming of the past 50 years is likely (>66%) to be attributable to human activities."

IPCC (2007) "Warming is unequivocal, and most of the warming of the past 50 years is very likely (90%) due to increases in greenhouse gases."

IPCC (2013) "It is still warming, greenhouse gases are **very likely** (>90%) responsible for most of the observed warming or 'virtually certain' (>99%)?

Regional details emphasizing what is or is not known

Message 4:
Temperature changes in the past few decades have exceeded those observed over 100,000s years and can be explained only by natural cycles AND human activity

A pause for messages from the skeptics

"All this talk about global warming... pah! I mean, where's the evidence?"

What is to come?

Issues to be covered

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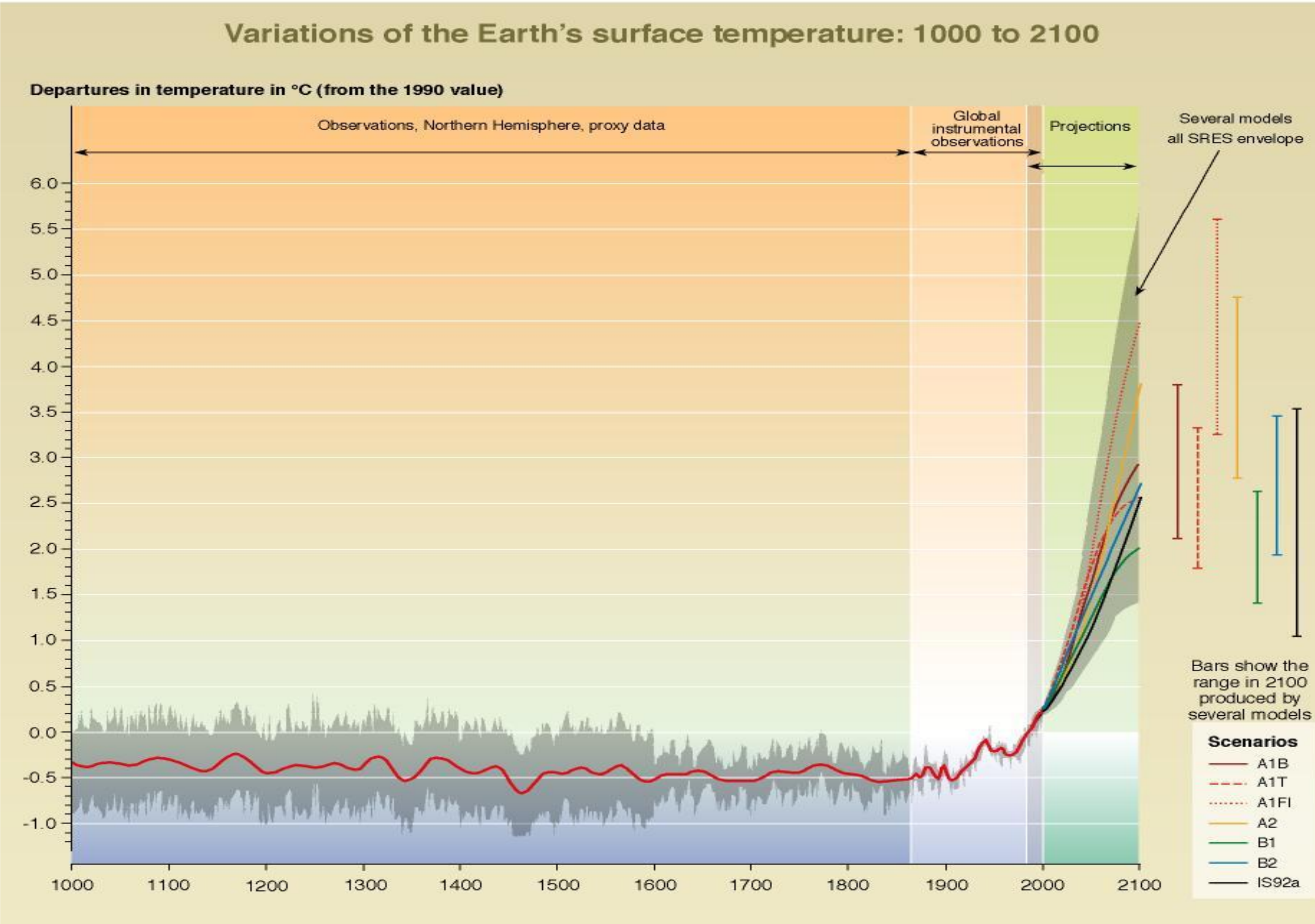
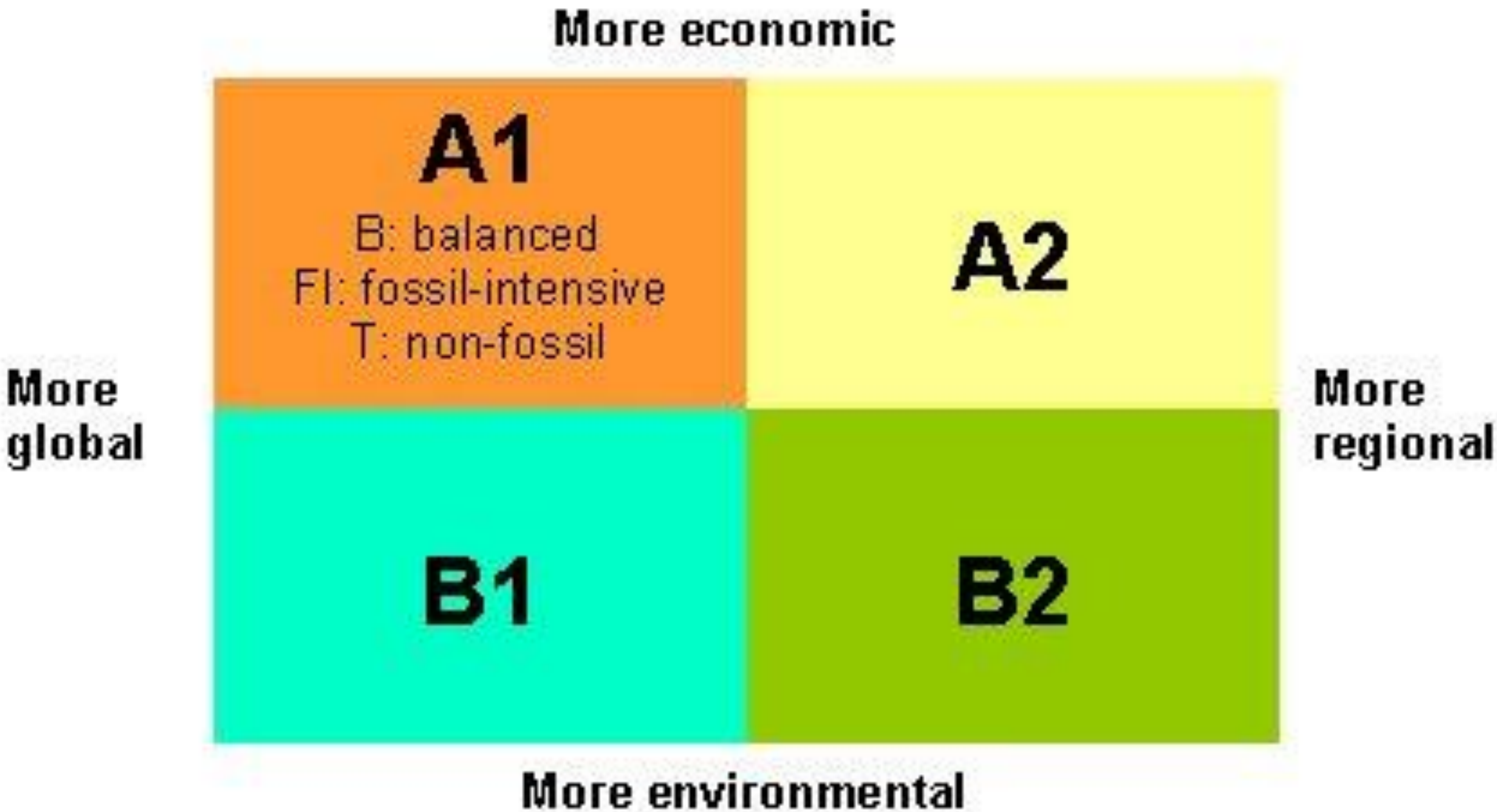
Important similarities and differences between DRR & CCA

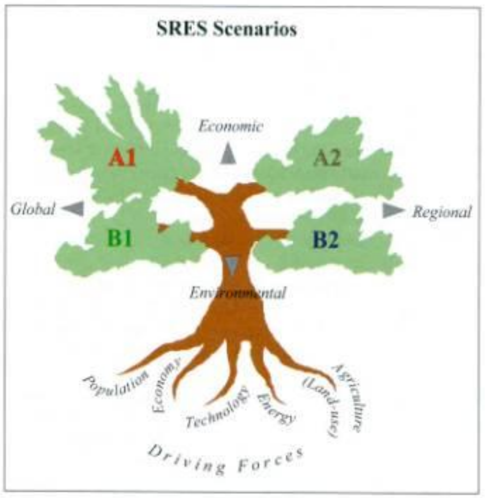
We often hear of the great uncertainty in climate projections

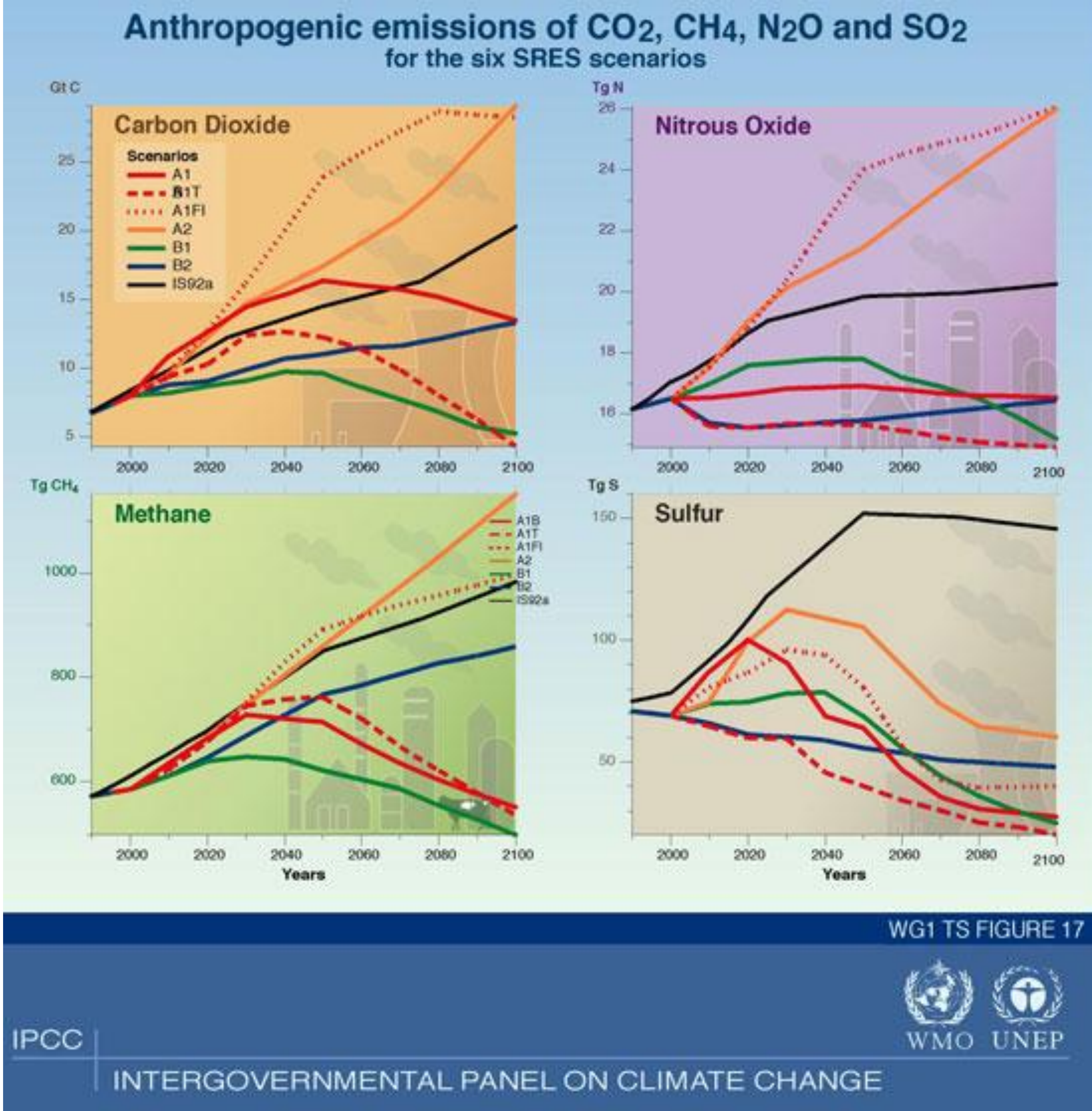
SPM - 10b

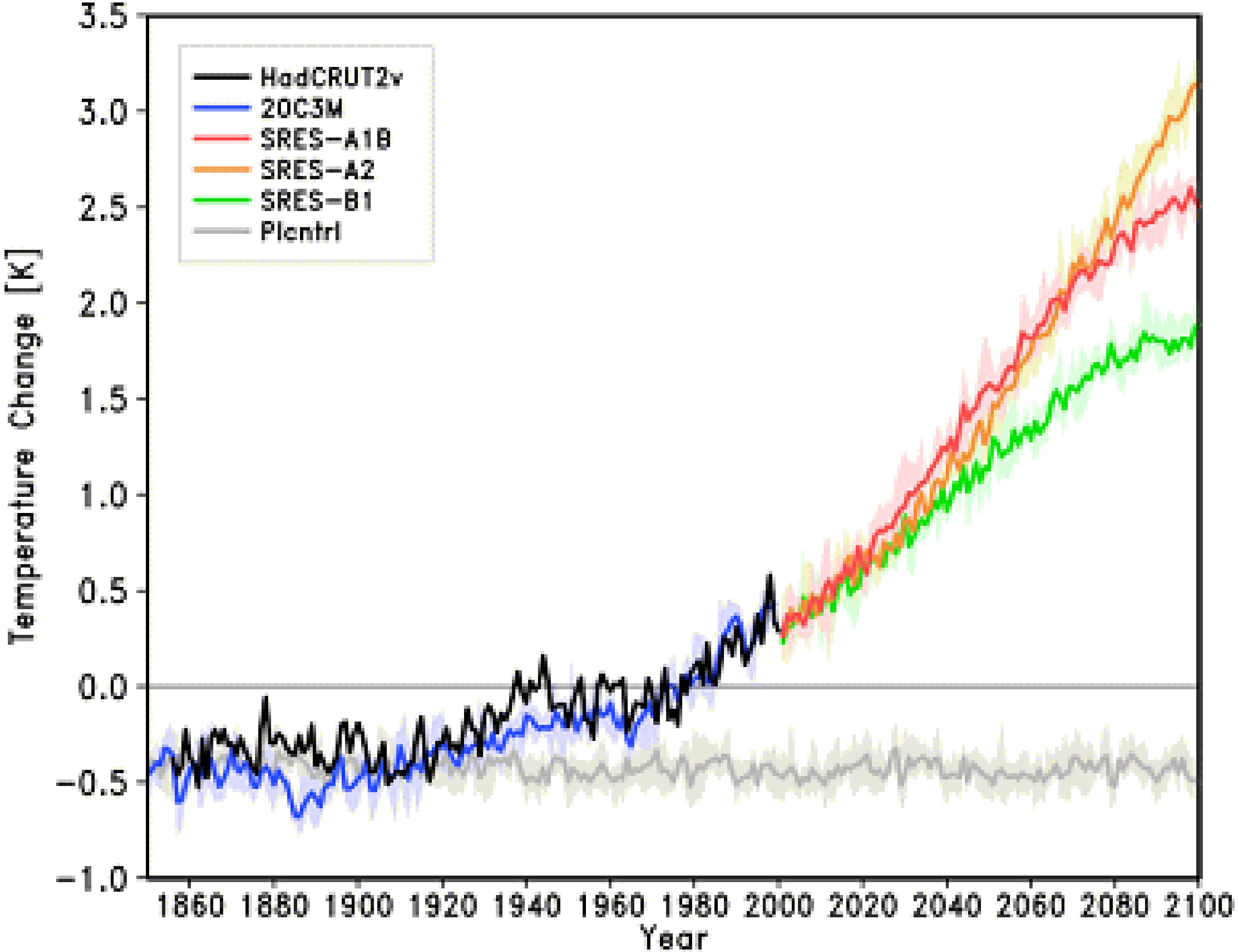



More economic
A1
B: balanced
FI: fossil-intensive
T: non-fossil
A2
More global
More regional
B1
B2
More environmental

IPCC SRES Scenarios

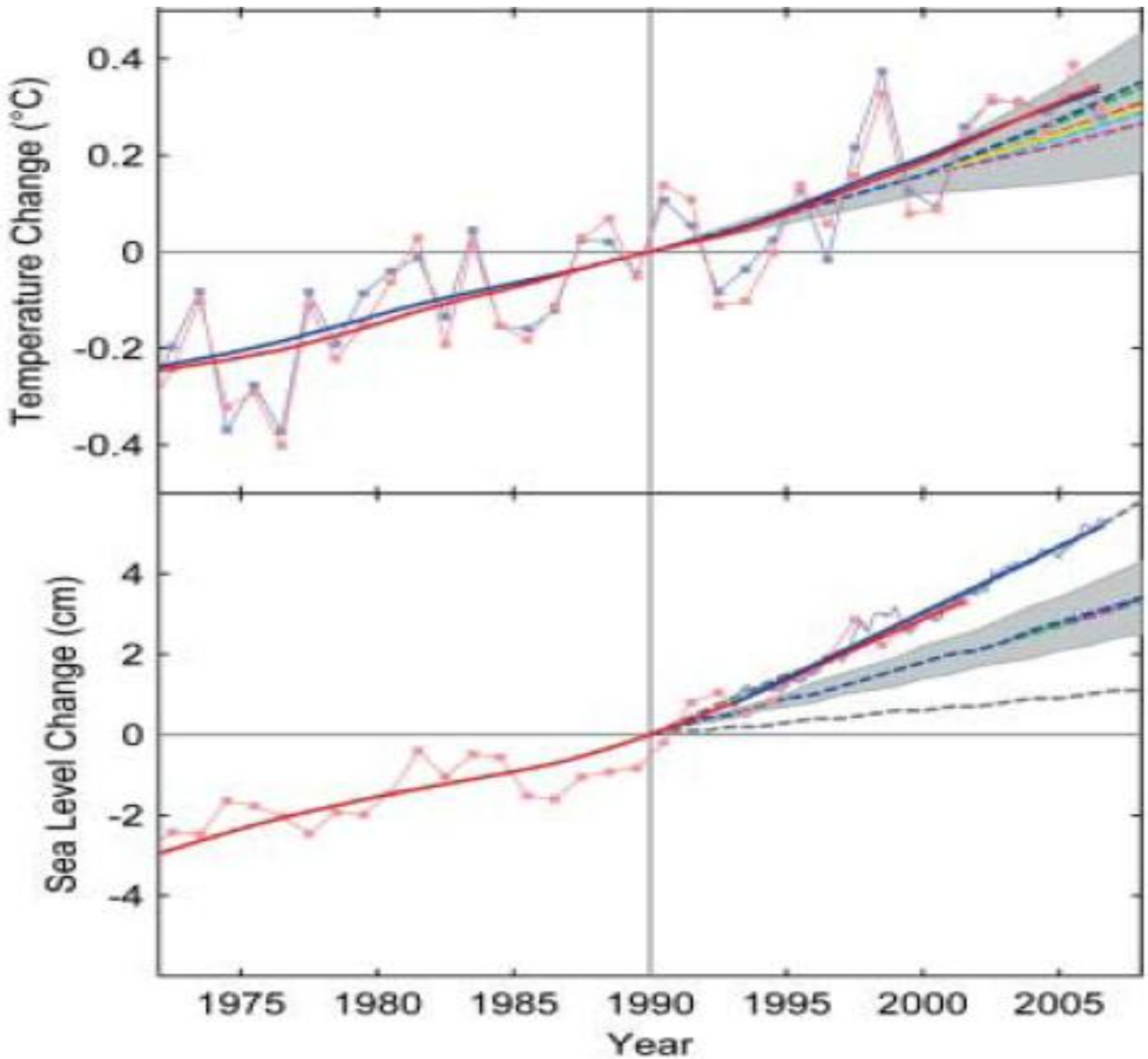





HadCRUT2v
20C3M
SRES-A1B
SRES-A2
SRES-B1
PIcntrl
Temperature Change [K]
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
1860
1880
1900
1920
1940
1960
1980
2000
2020
2040
2060
2080
2100
Year

Message 5:
Some uncertainty in climate projections arises from the models themselves; most arises from our uncertainty as to what emissions pathway we will choose to follow

Observed change are ahead of models & scenarios
Rahmstorf et al Science 2007



The 'smooth' lines represent the different scenarios of future climate as often used in the IPCC.

The lines with the dots show the summary of observed trends by two independent organizations.

Observations are tracking or exceeding the most extreme scenarios.

Message 5a:
Our current emissions pathway is leading to observed changes that exceed the most pessimistic of the SRES scenarios

What do these projections mean for DRM?

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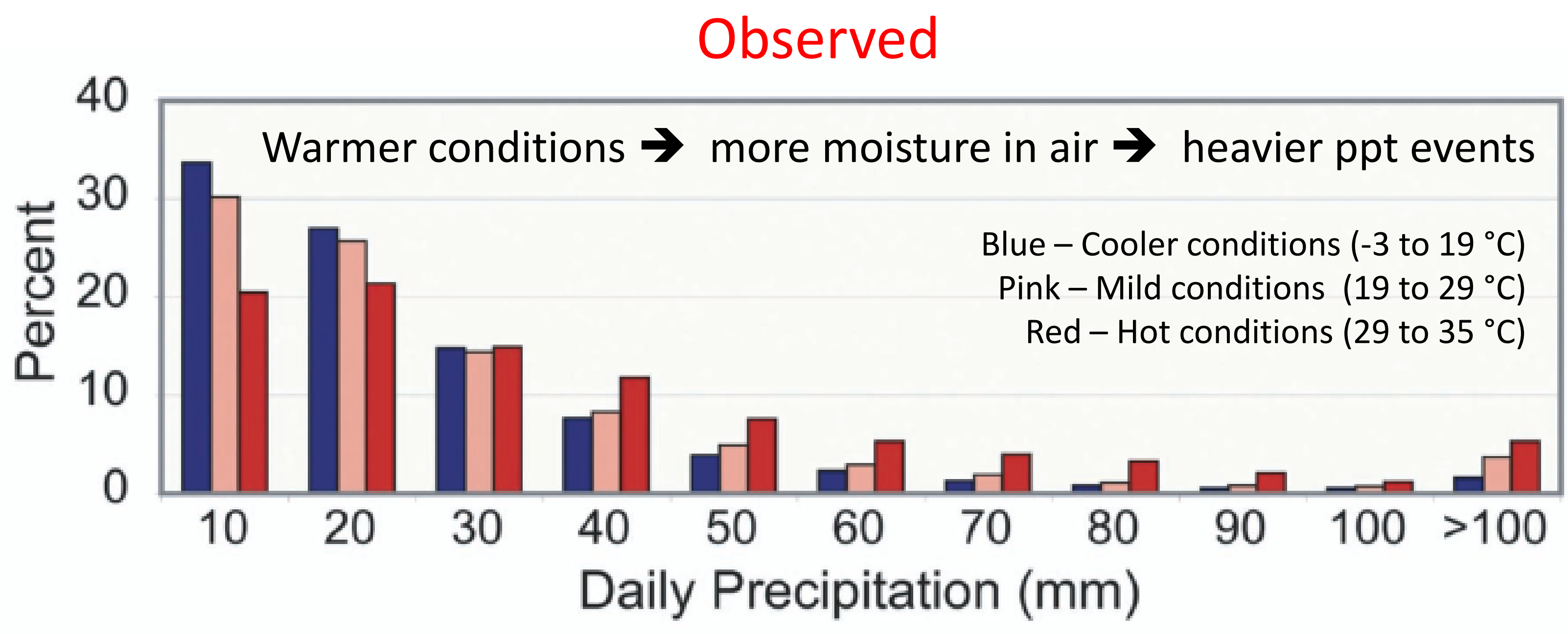
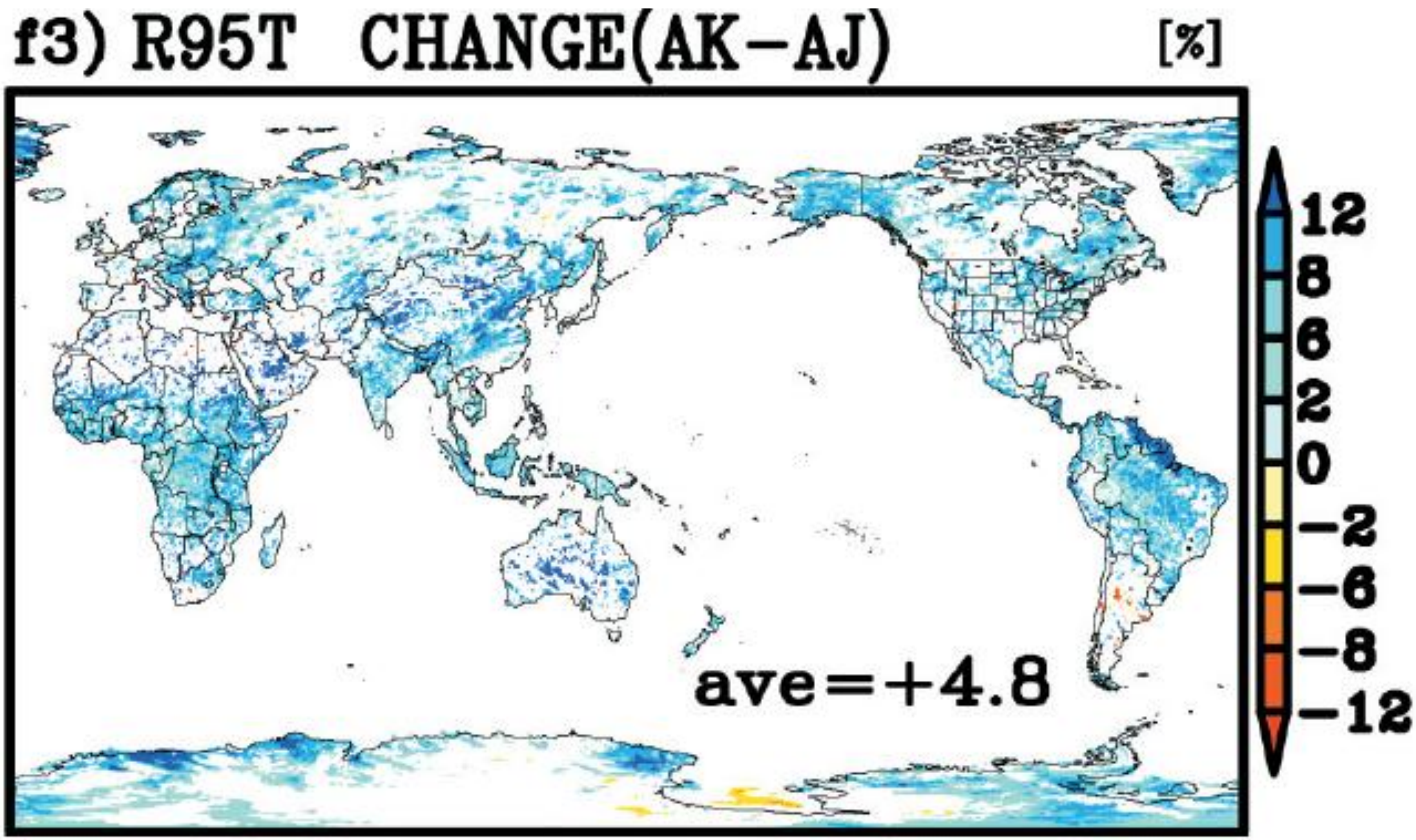
Important similarities and differences between DRR & CCA

In a warmer world precipitation will come in more extreme events; i.e. mm rain per day or per hour will increase



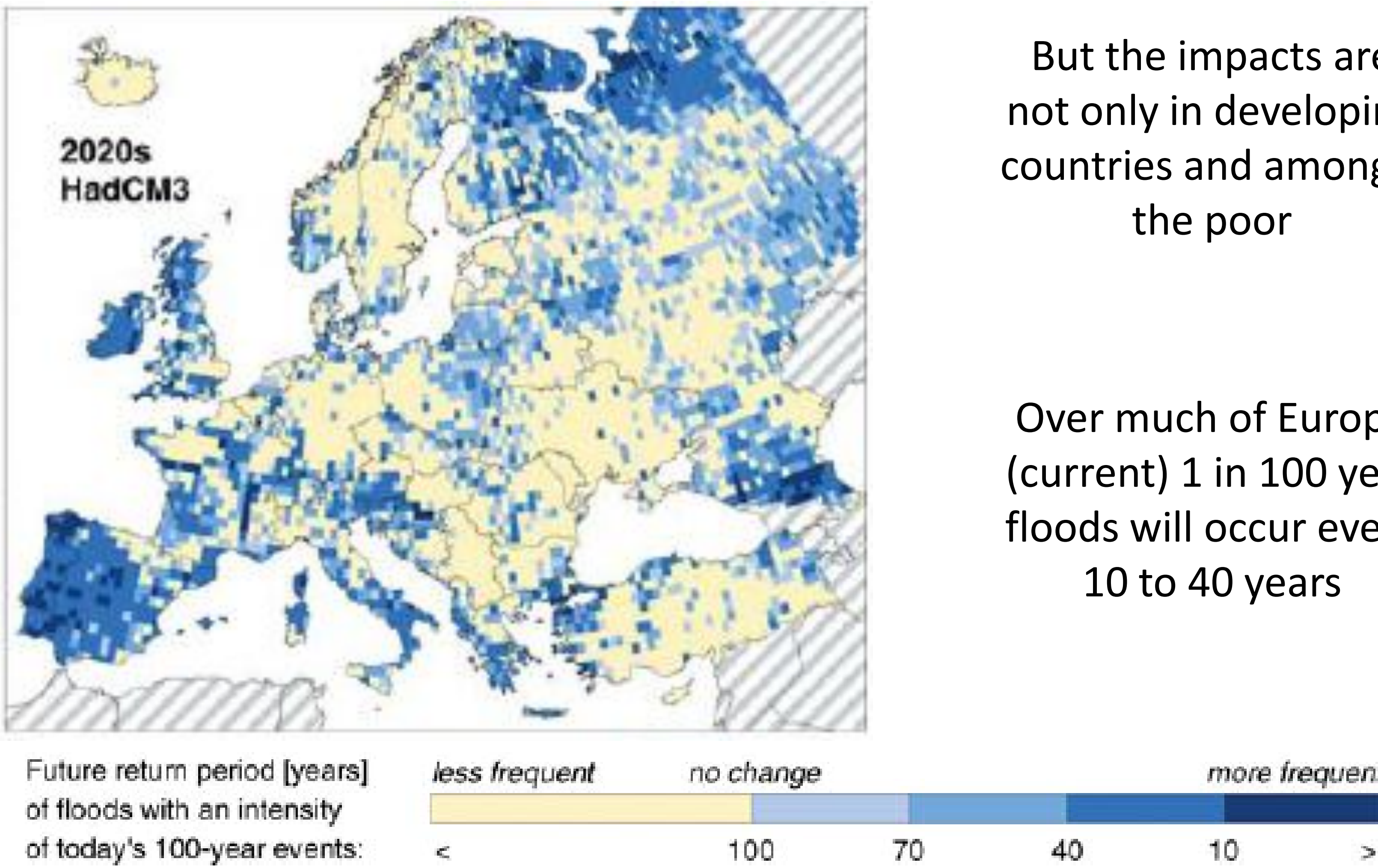
Model outputs show this same trend

Percent change in annual rainfall falling in extreme events (>95% of current intensity range)



100-year flood frequency 2020

Lenher et al 2006 Climatic Change

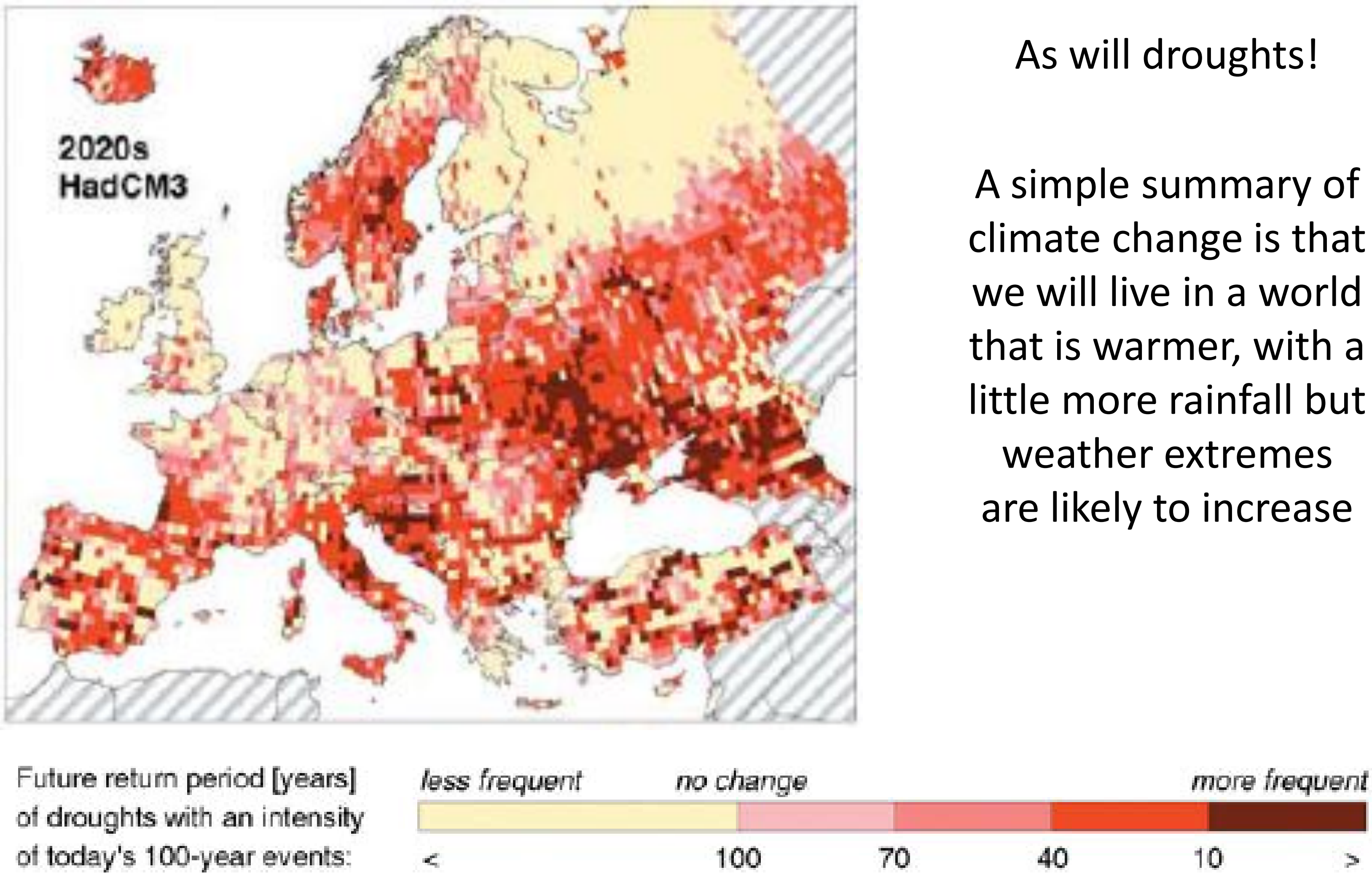


But the impacts are not only in developing countries and amongst the poor

Over much of Europe (current) 1 in 100 year floods will occur every 10 to 40 years

100-drought frequency 2020

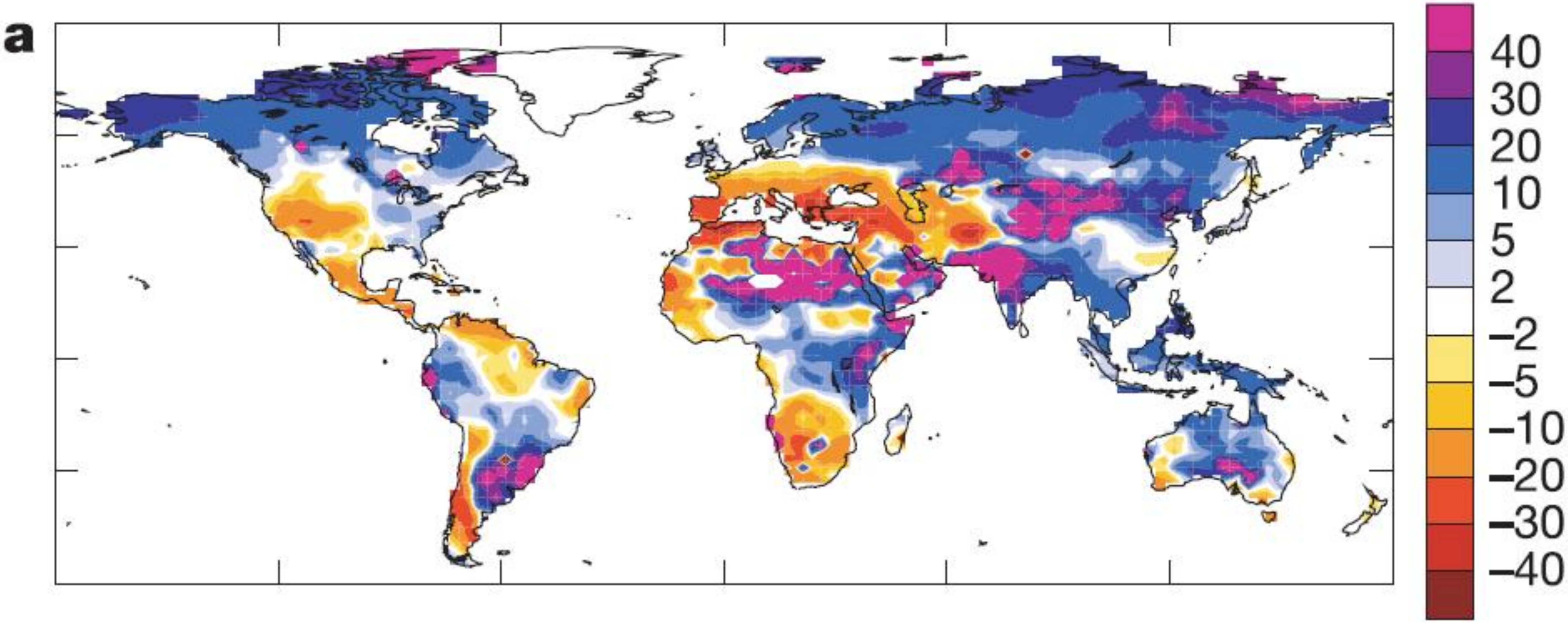
Lenher et al 2006 Climatic Change



As will droughts!

A simple summary of climate change is that we will live in a world that is warmer, with a little more rainfall but weather extremes are likely to increase

Modelled - % change in runoff by 2050

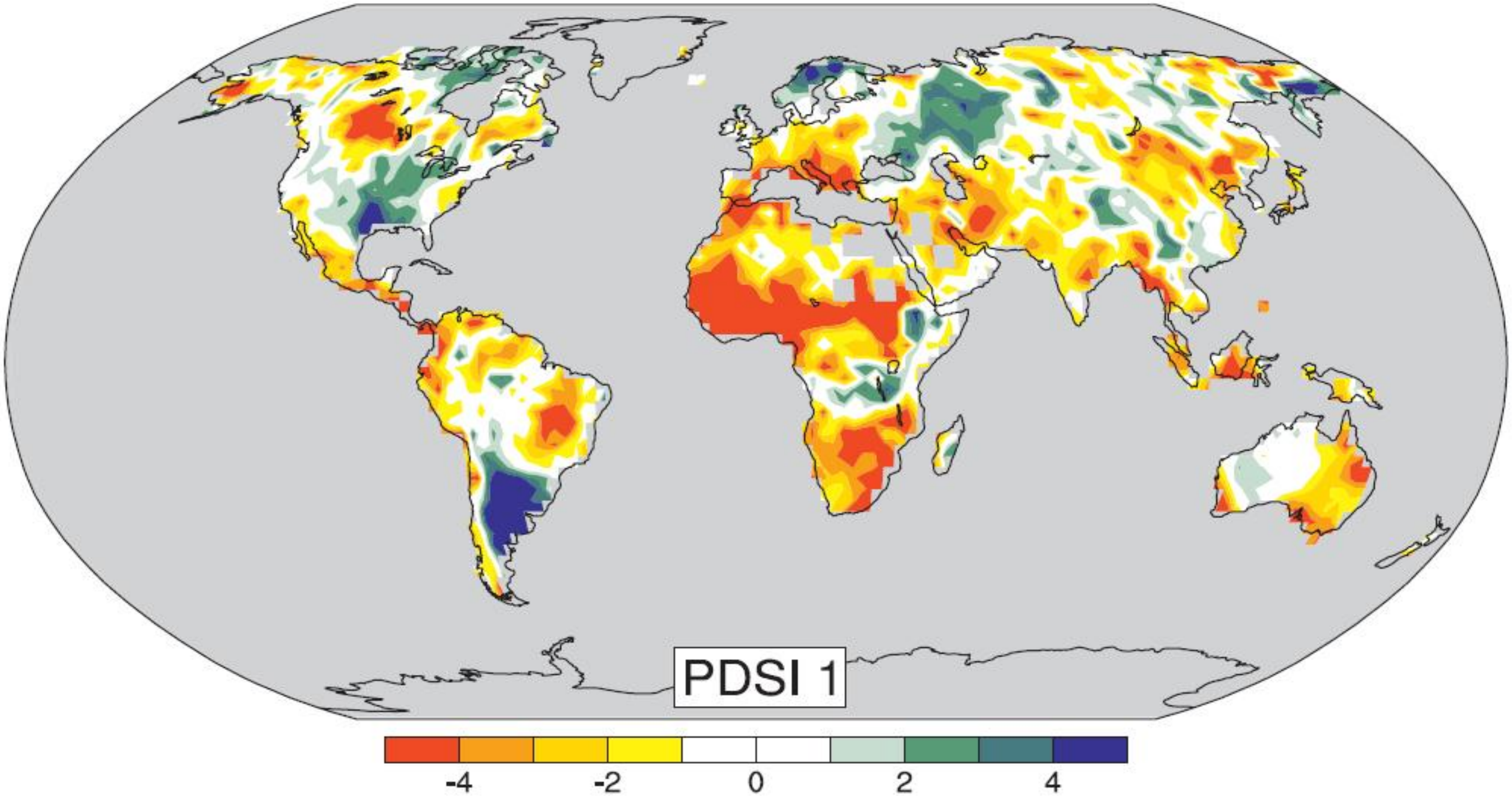


Many of the major “food-bowls” of the world are projected to become significantly drier

Milly et al Nature 2005

Observed -- Palmer Drought Index 20th Century trends

Q&A SAR4 WG1



Projected - Changes in cereal production

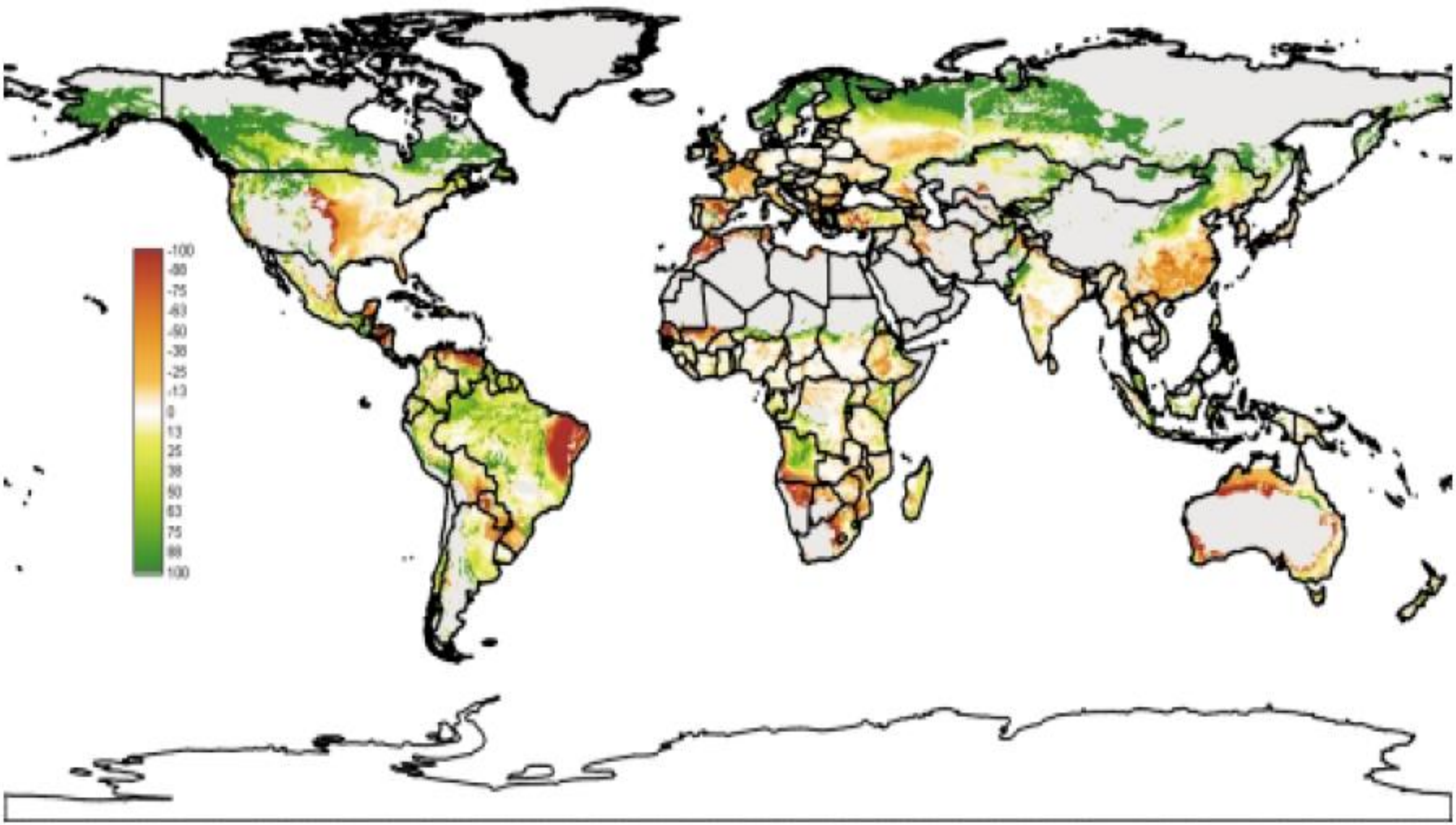
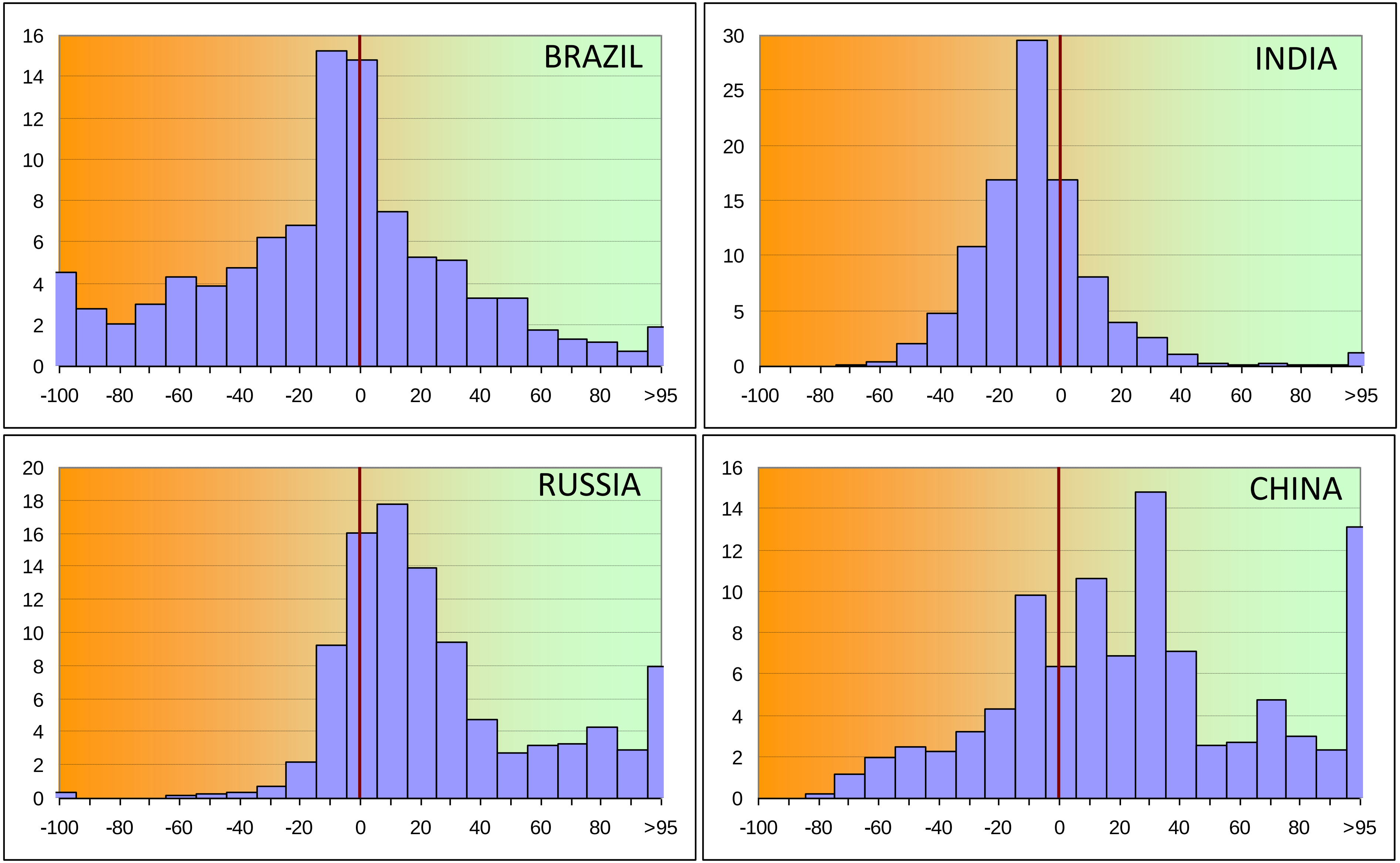


Plate 3.12b. Change in suitability for rain-fed cereals (Had3-A1FI, 2080s).

IIASA Jo'burg 2002

Proejected - Countries are affected differently; some are advantaged

Climate impacts on cereal production capacity, ECHAM4 2080s, Rain-fed multiple cropping



But even a projected benefit comes with social costs and trade impacts

From Liu et al

2005

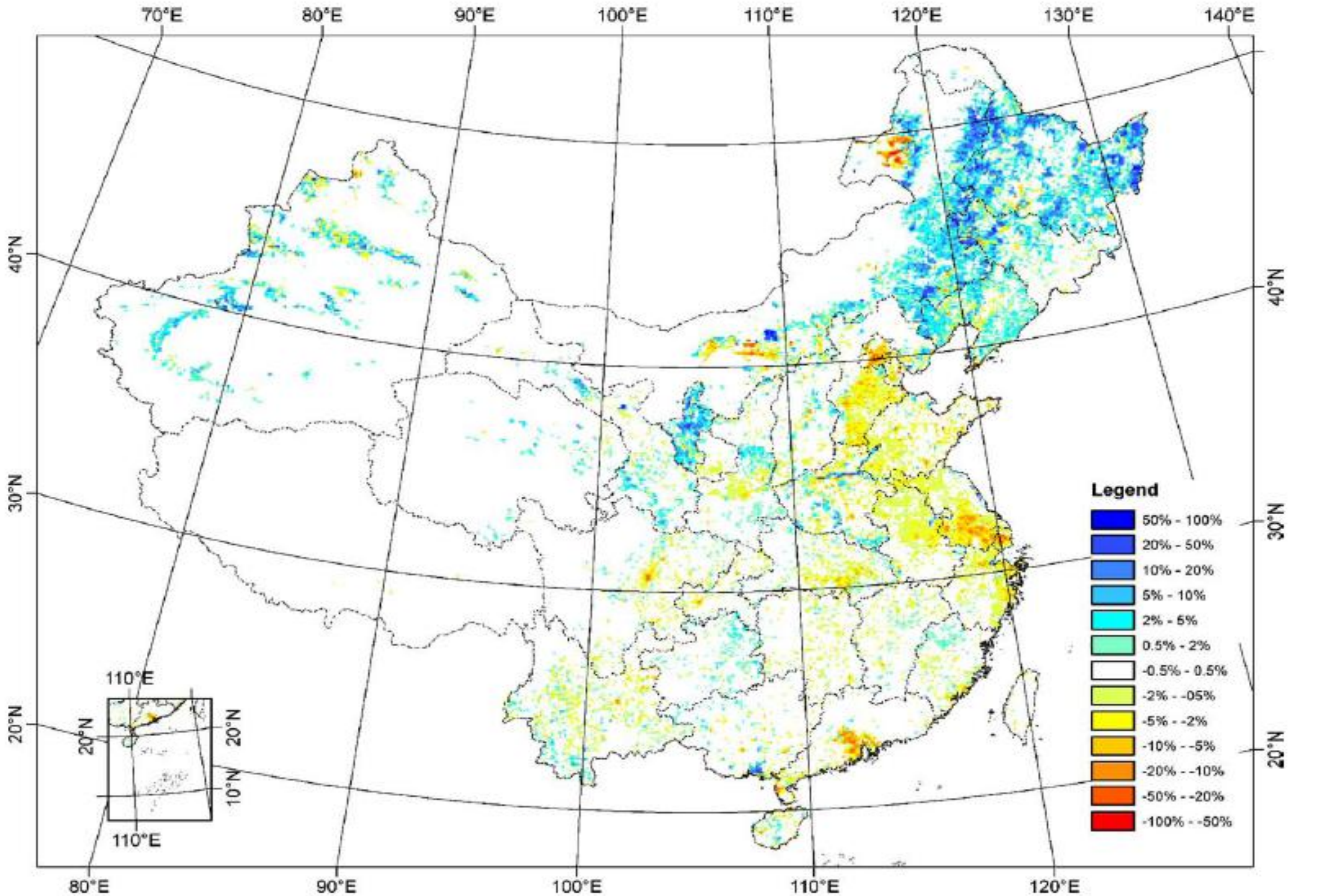
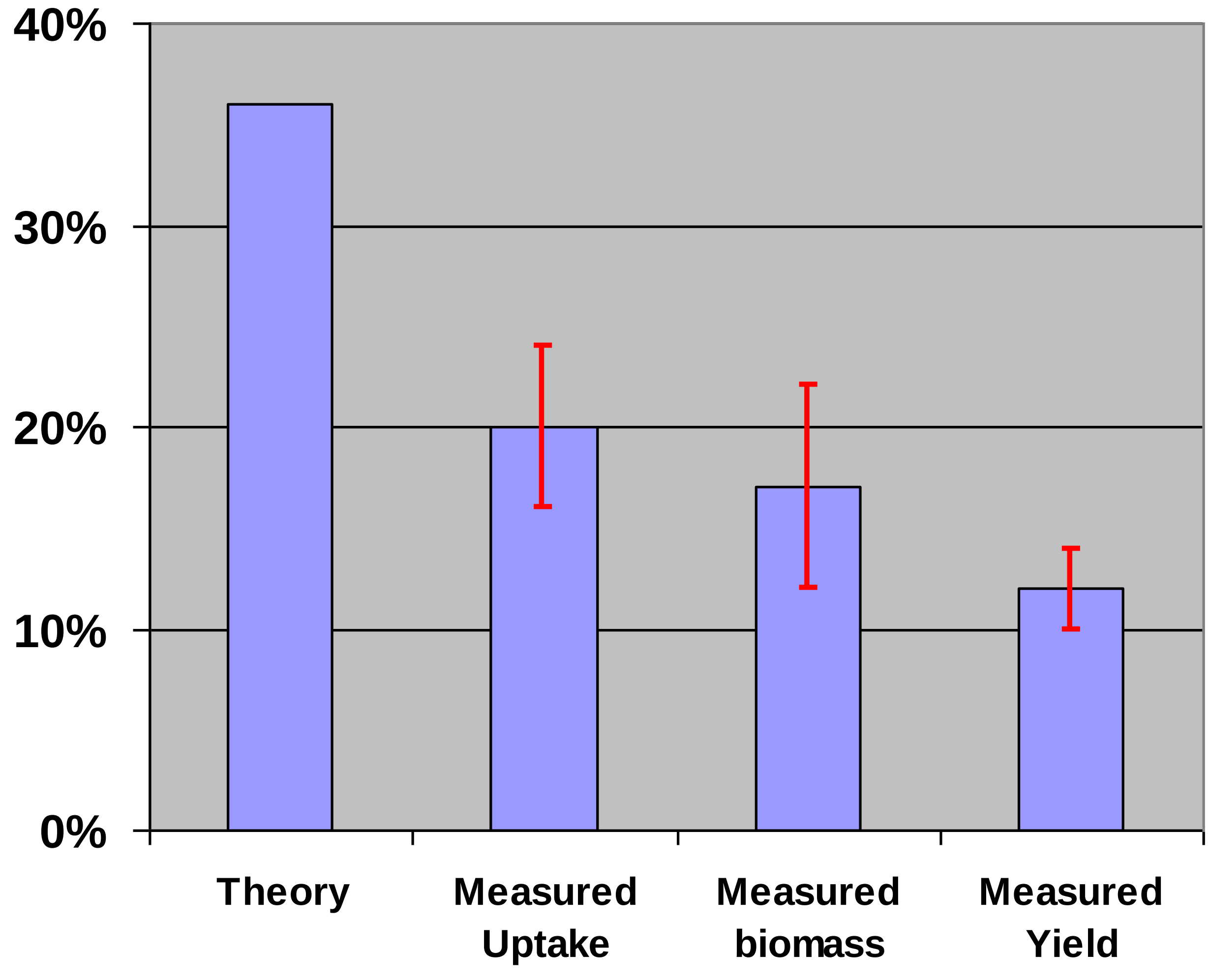


Fig. 6. The spatial distribution of cropland change during 1990 – 2000 (units: percentage). The value shows the fraction of cropland change in each grid during this period, e.g., 50% means 50% of grid area (i.e. 0.50×1 km^2 = 50 ha of cropland) has changed in each grid. The cropland increase is represented by blue color and the cropland decrease is represented by red color.

CO_2 Fertilisation Effect

Higher CO2 levels in the atmosphere, in theory makes plant growth more efficient – i.e. faster and using less water



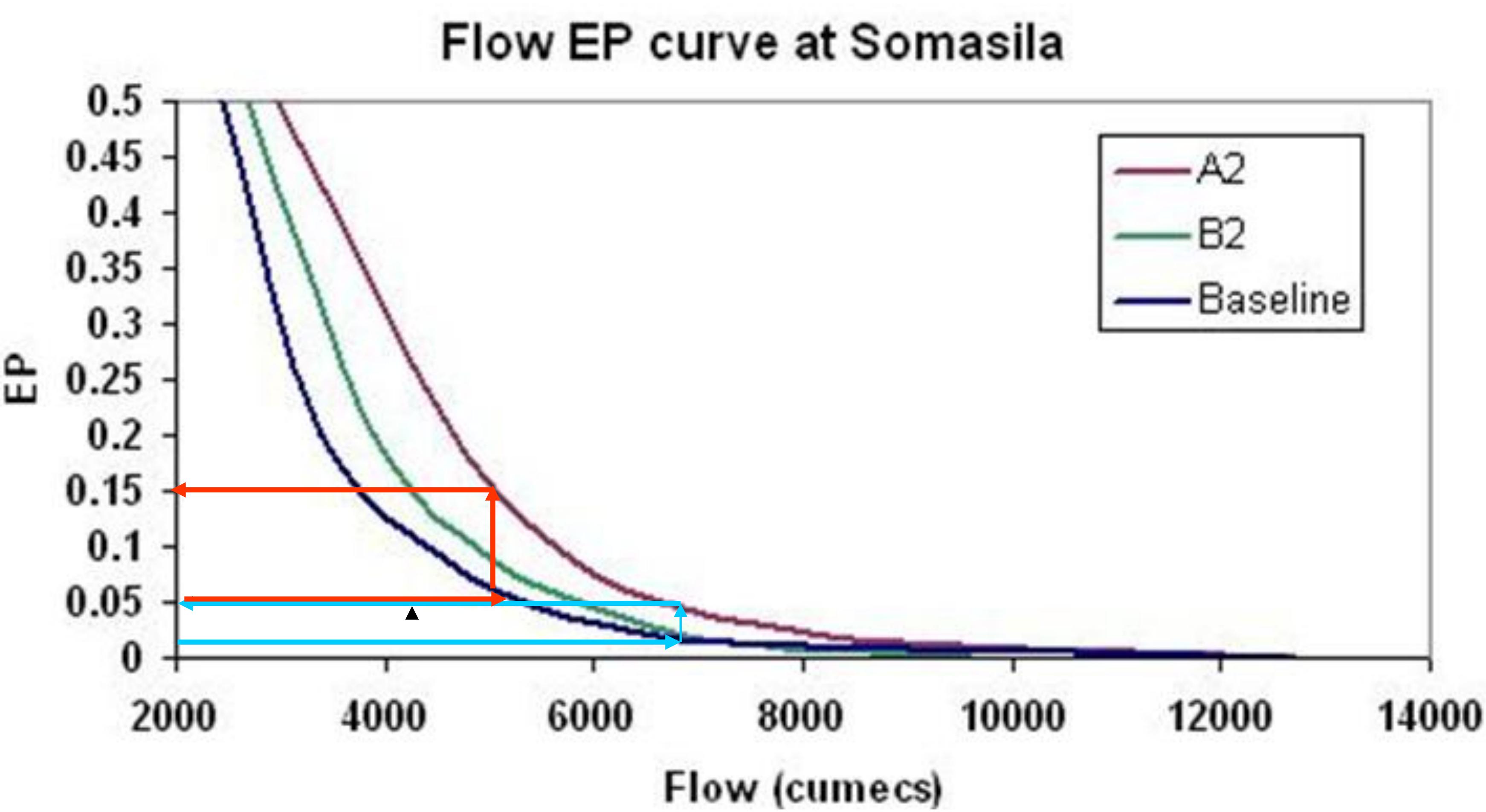
From Long et al 2007 Science 312, 1918-21

Message 6:
Many parts of the world, and especially some of the main food producing areas, will be effectively drier but with more floods.

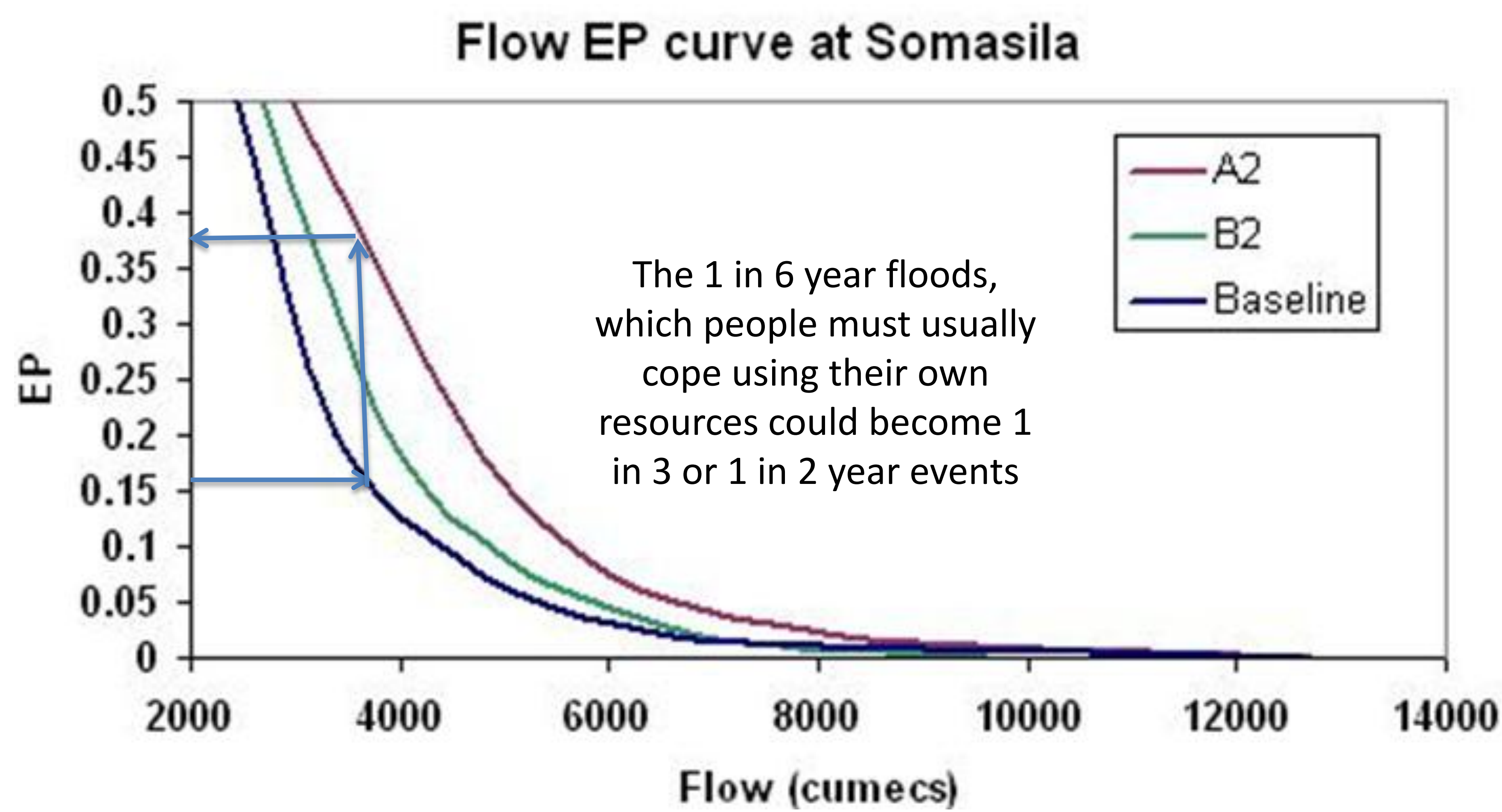
Even modest changes in rainfall events can dramatically affect flood frequencies

1 In 100 year flood returns every 20 years

1 in 20 year flood returns every 6 years



Not only will "disastrous" flood events increase, but so too will smaller floods that cause chronic losses to families and their livelihoods



Those “bad events” add up

Message 7:
Not only will "disastrous" flood events increase, but so too will smaller floods that cause chronic losses to families and their livelihoods

The U.S. Climate Change Science Program

Observed

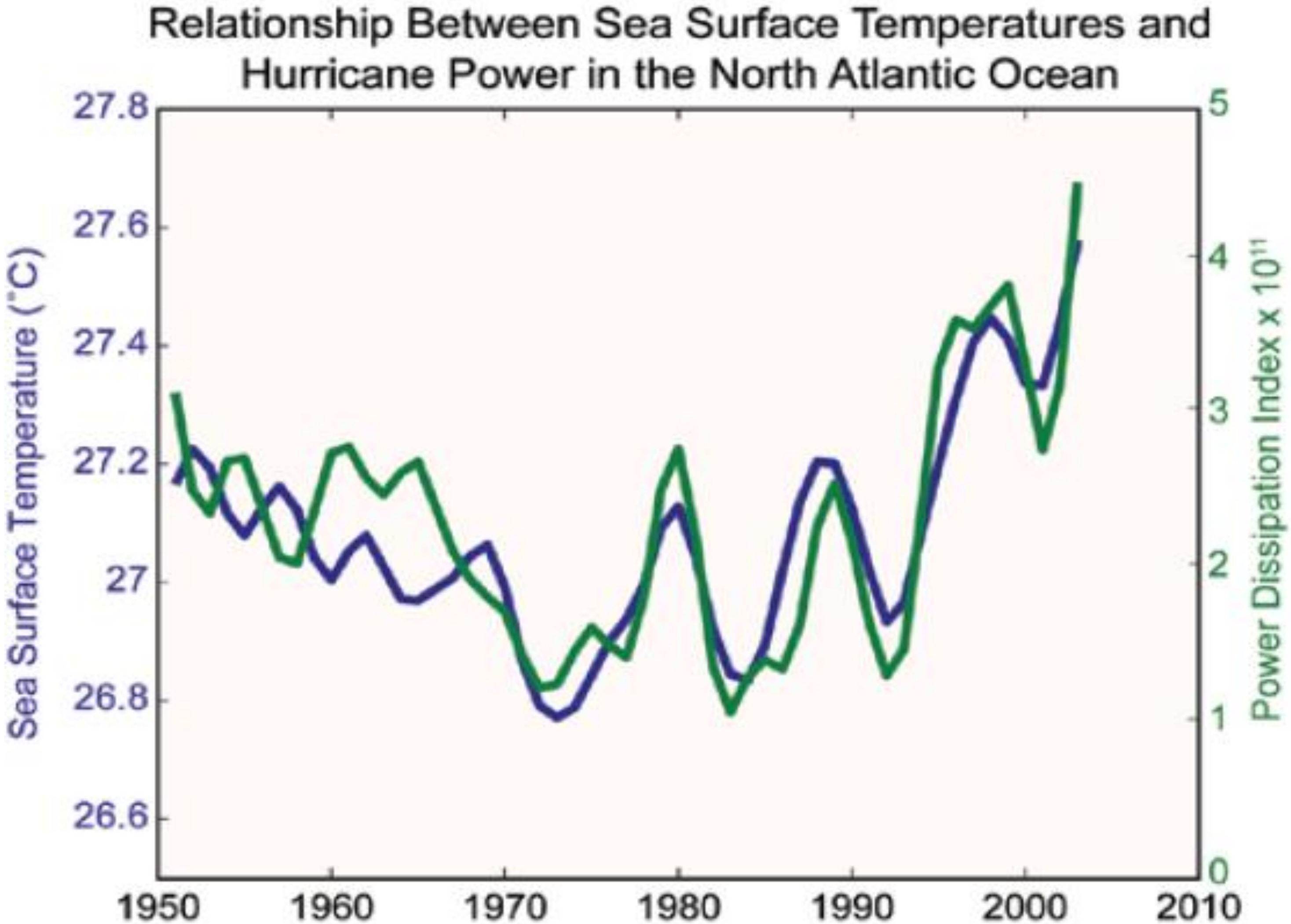
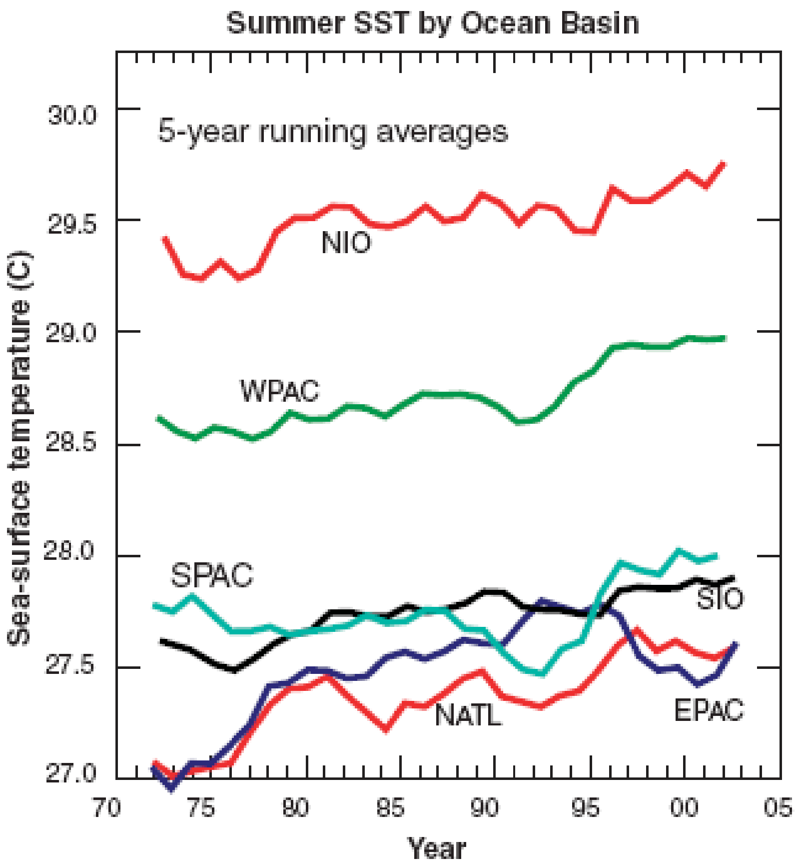
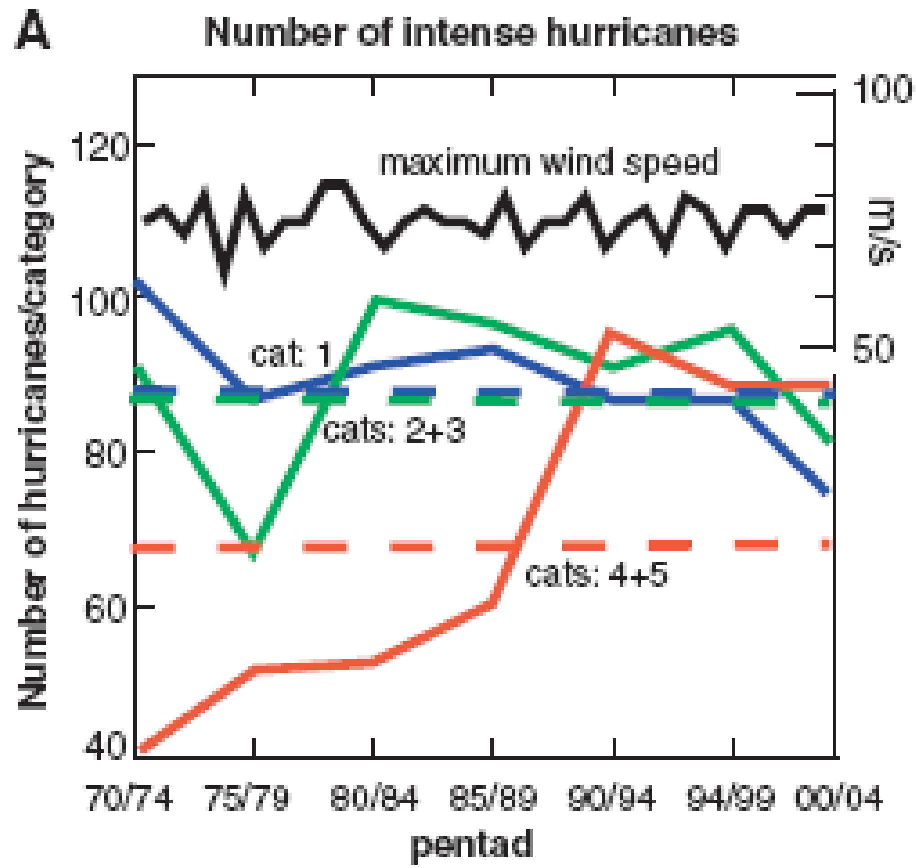


Figure ES.6 Sea surface temperatures (blue) and the Power Dissipation Index for North Atlantic hurricanes (Emanuel, 2007).

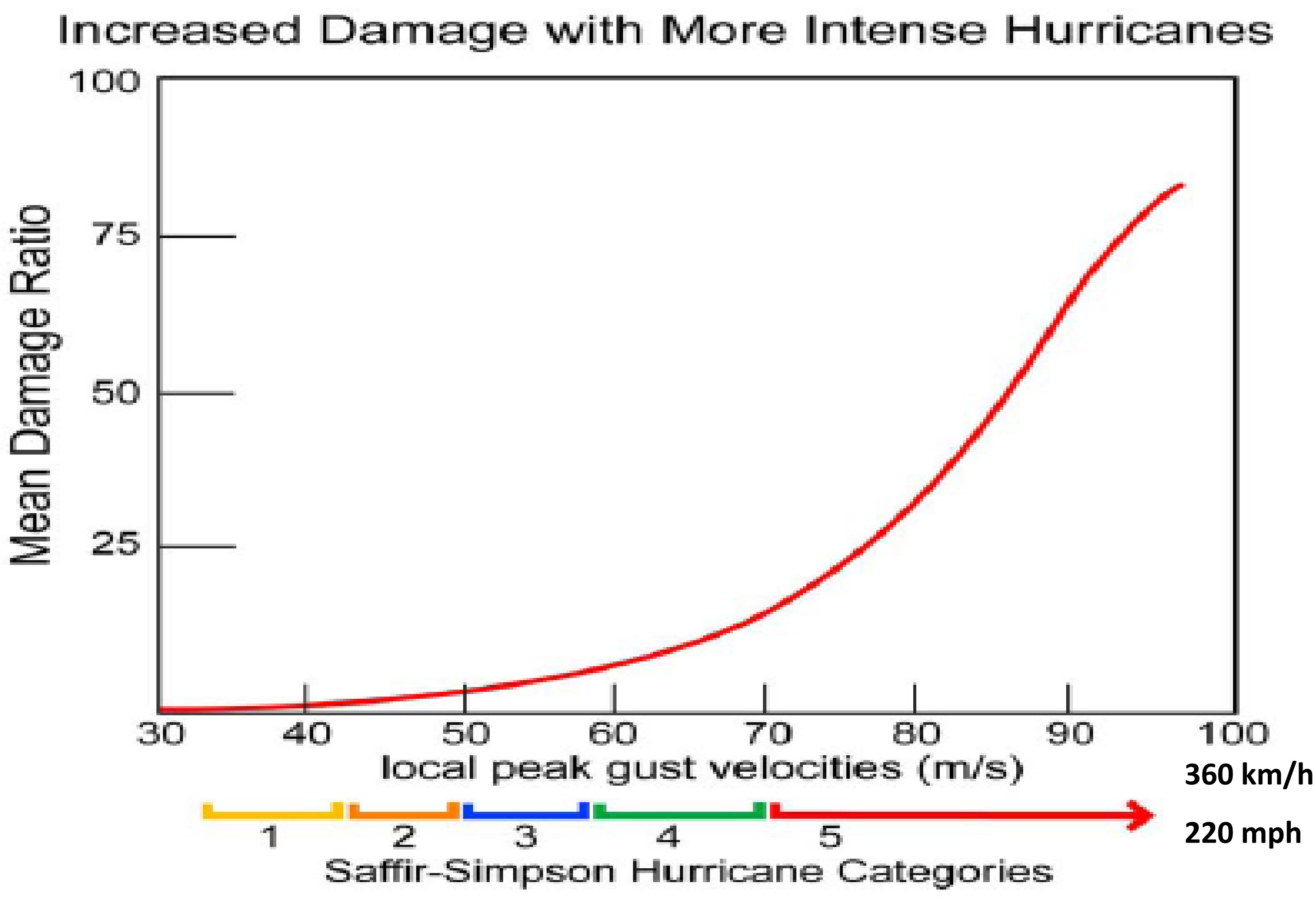
Cyclone intensity - Observed

Webster et al 2005 Science 309, 1844

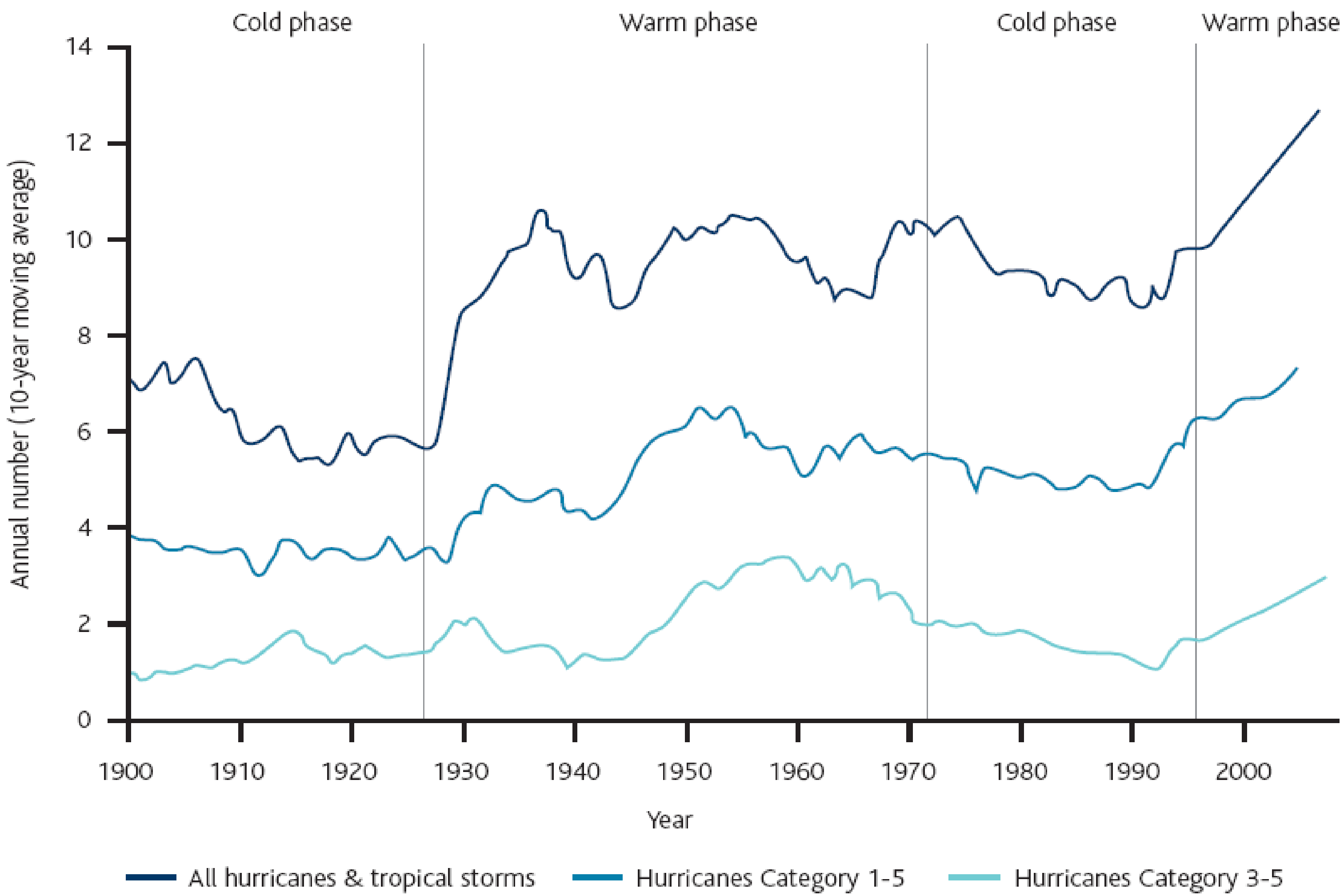




Meyer et al 1997



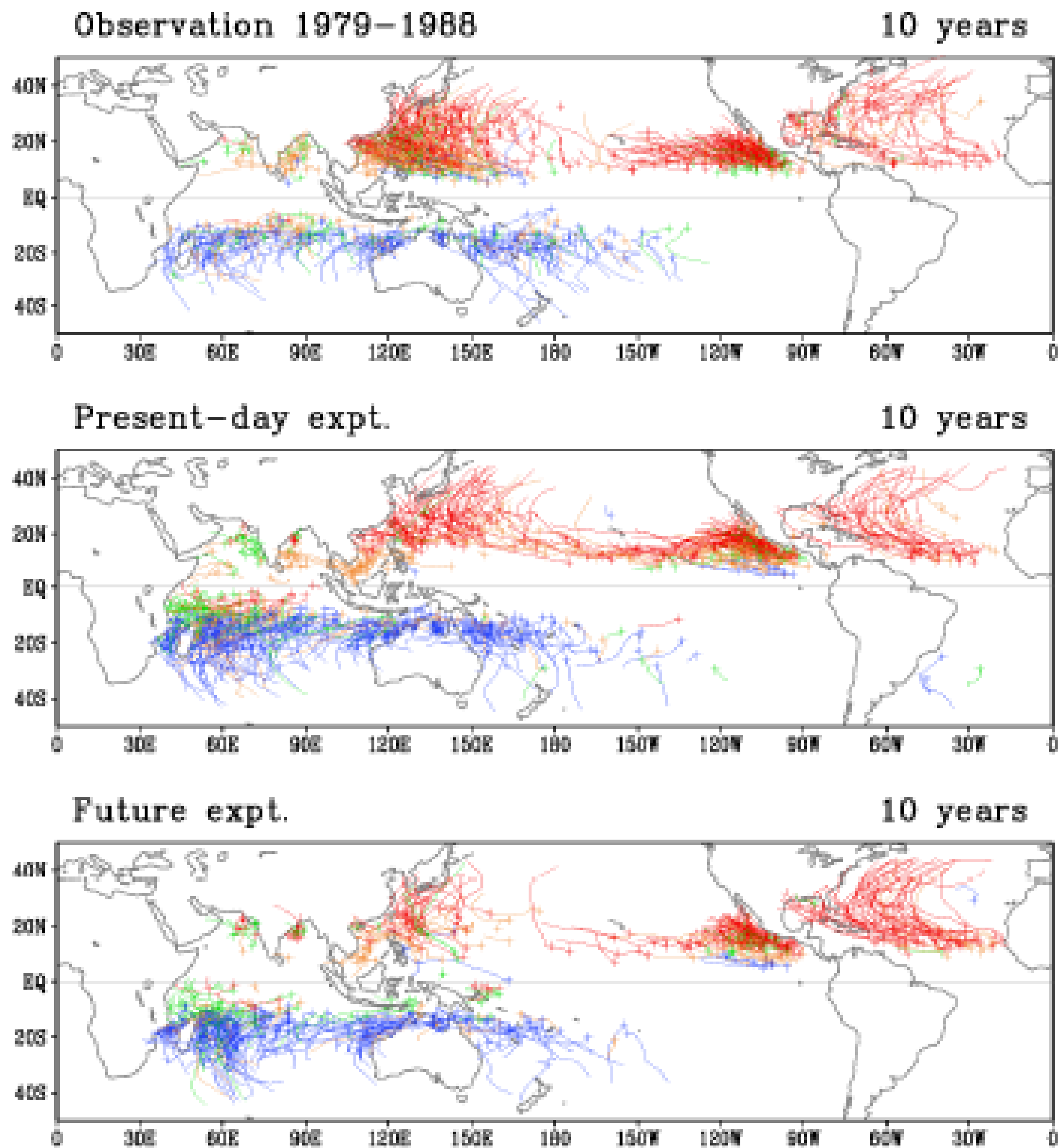
Observed



Source: NOAA, with re-handling and calculations by Munich Re.

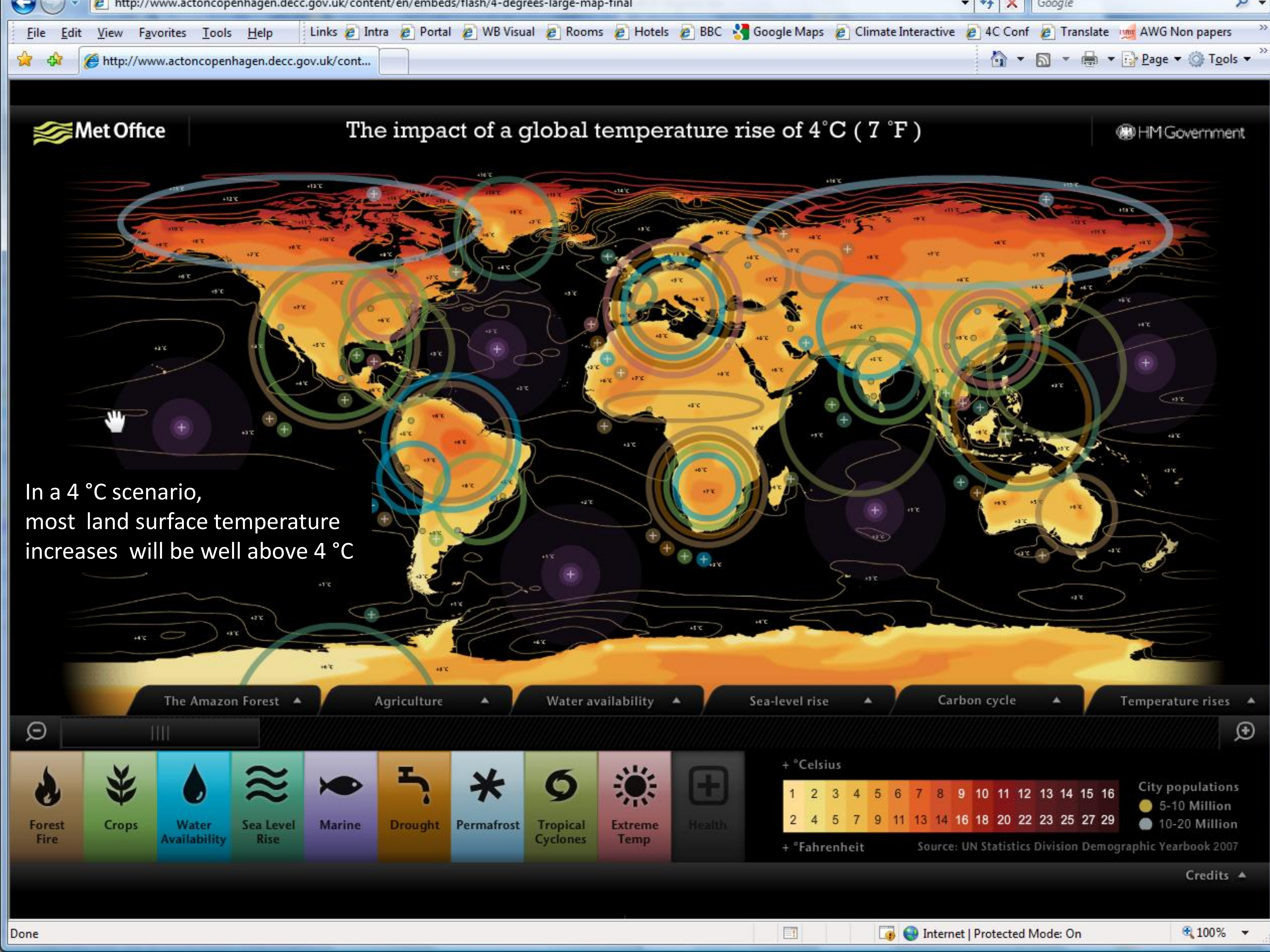
Note that recent cold and warm phases show higher frequencies than previous ones

- Noda et al Earth Simulator from Oouchi et al 2005
- Cyclone frequency at end of 21st C
- Overall frequency will fall but number of intense events will increase



But don’t expect much detail on cyclone paths

Message 8:
A simple summary of climate change is that we will live in a world that is hotter, with a little more rainfall, but weather extremes, wet, dry and stormy, are all likely to increase



In a 4 °C scenario,
most land surface temperature
increases will be well above 4 °C

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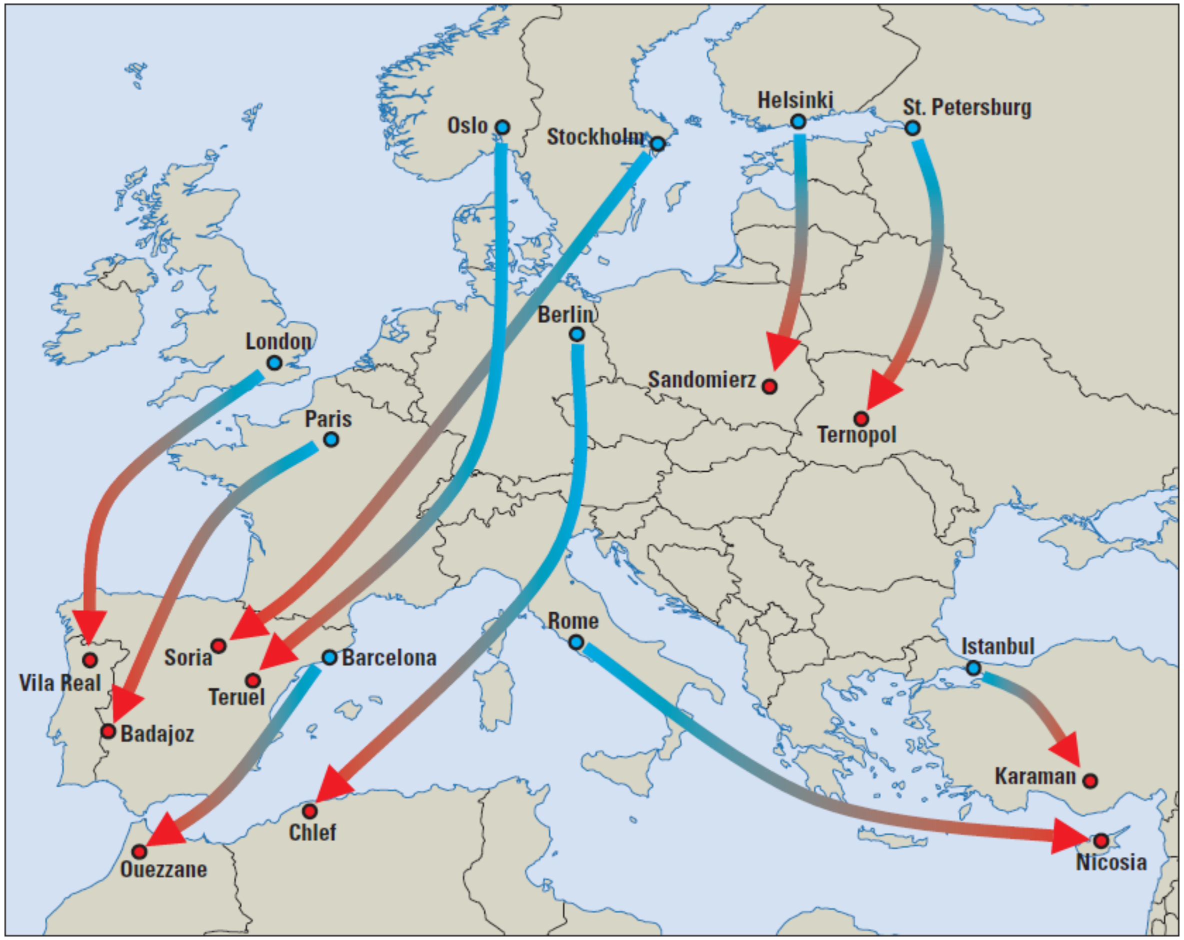
Global Facility for Disaster Reduction and Recovery

It is not all about extremes and disasters.

Imagine living through summer after summer in an 'Oslo' designed house but with the weather of central Spain

Late 21st Century projections

Map 2.3 Northern cities need to prepare for Mediterranean climate—now



Source: WDR team, reproduced from Kopf, Ha-Duong, and Hallegatte 2008.

Message 9:
The world is not homogenous

Global averages can be misleading and we must look at the regional & local effects

But can we get that resolution?

Limits to what we can know and the downscaling issue

Issues to be covered

Is the changing climate affecting disaster risk and risk management?

What can we learn from climate science and climate modelling?

What is to come?

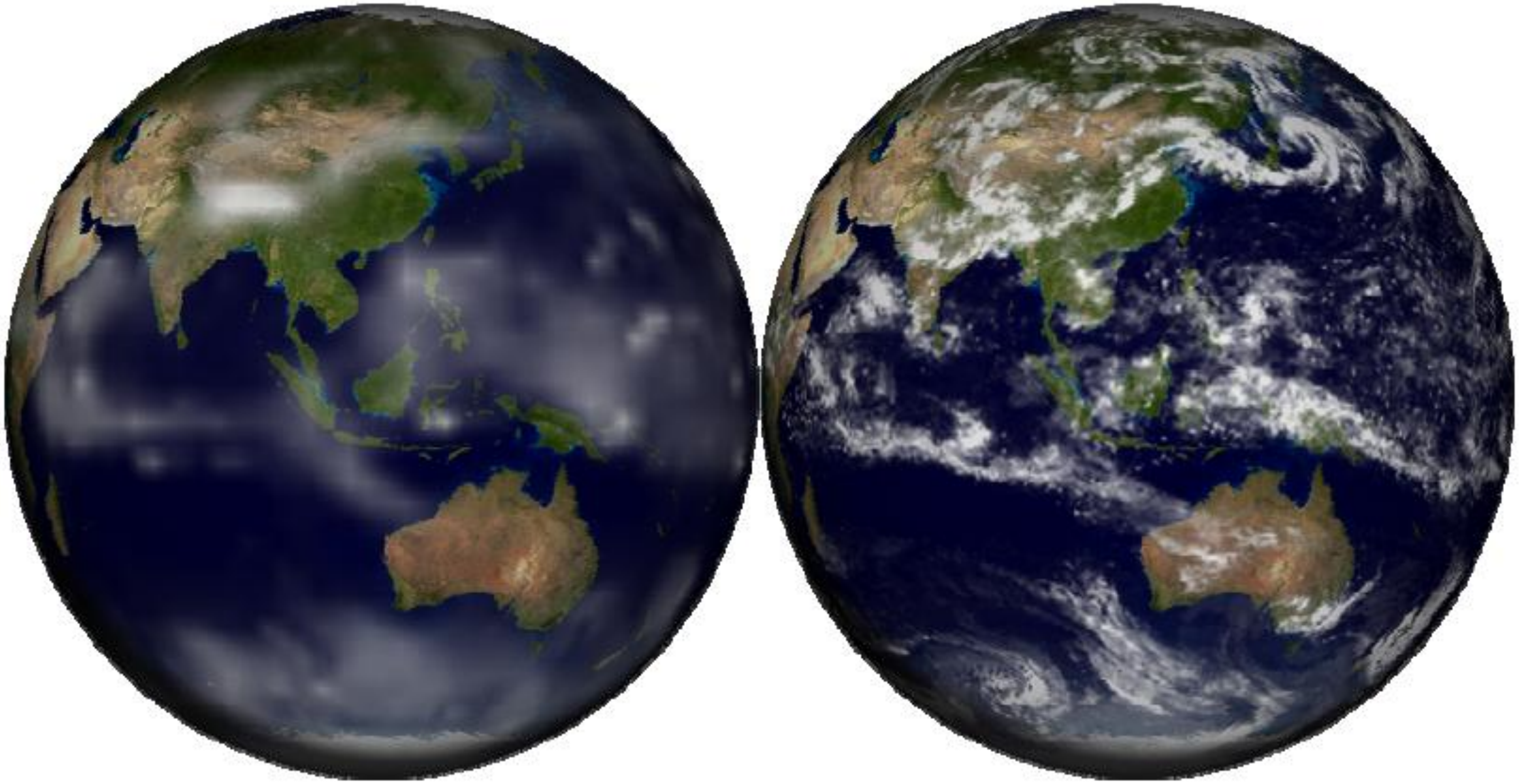
What do these projections mean for DRM?

Limits to what we can know and the downscaling issue

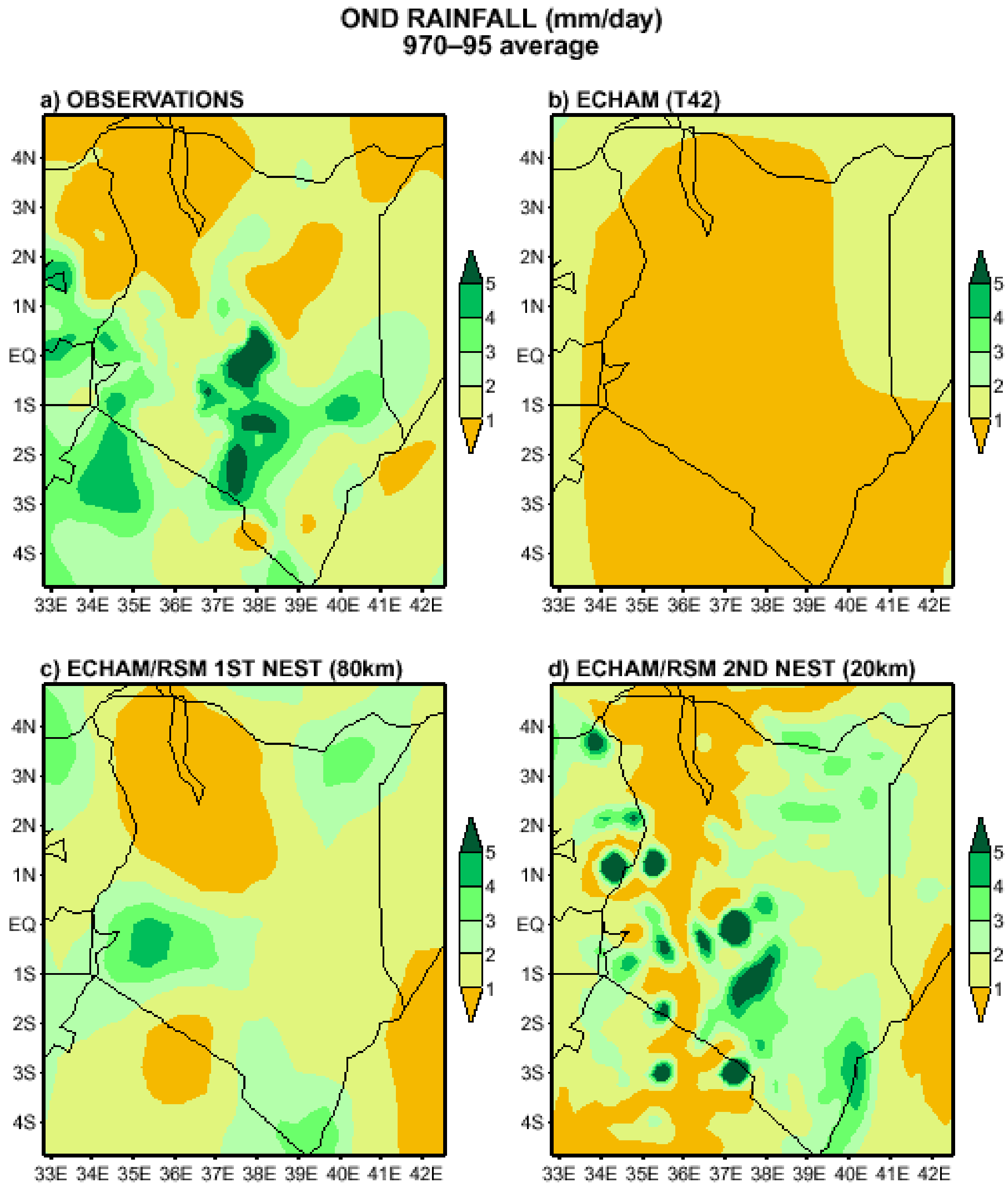
Important similarities and differences between DRR & CCA

Takeshi Enomoto, Earth Simulator Center/JAMTEC

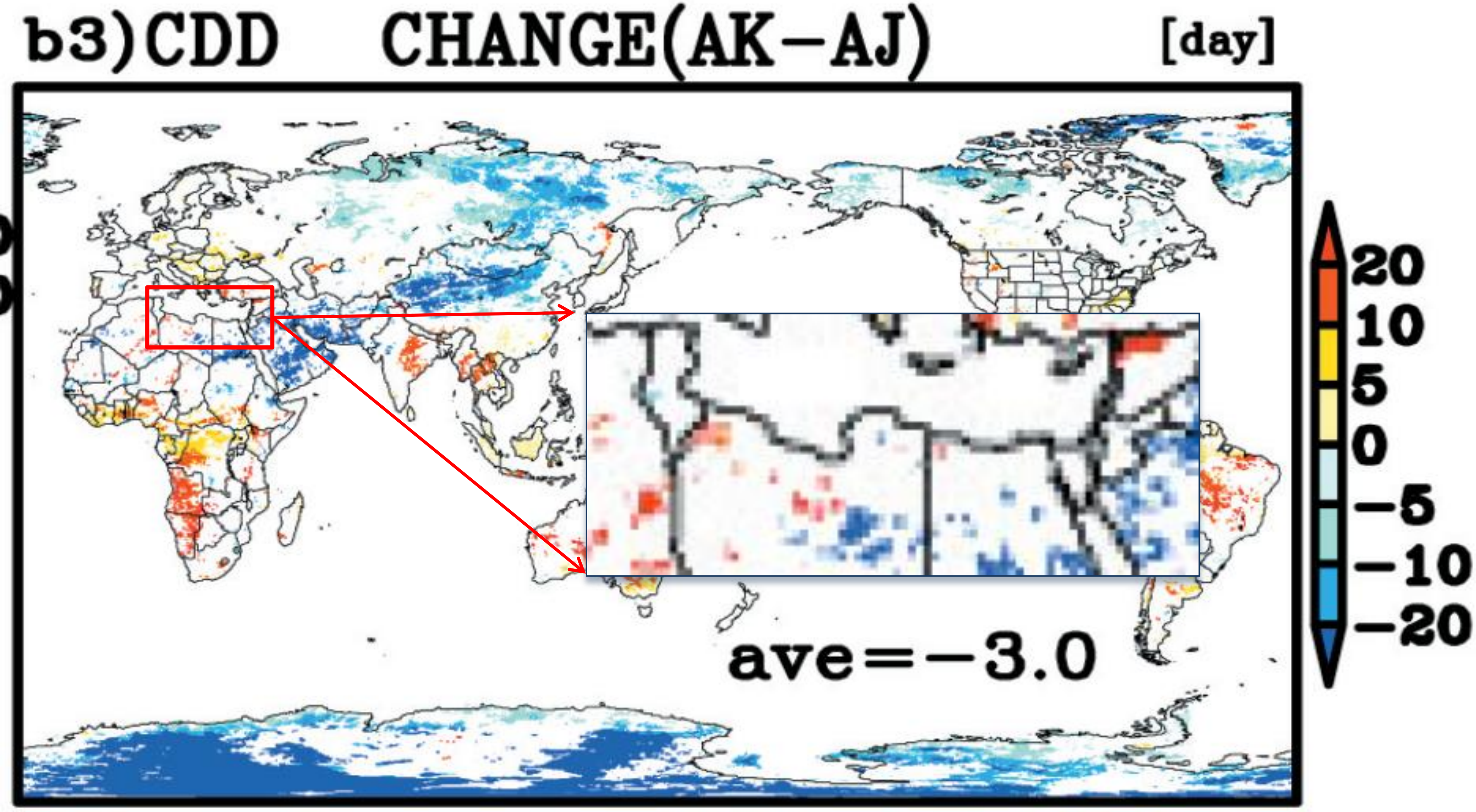
Cloud representation at 320 km grid (Left) versus 20 km (Right) for the same point in a model run



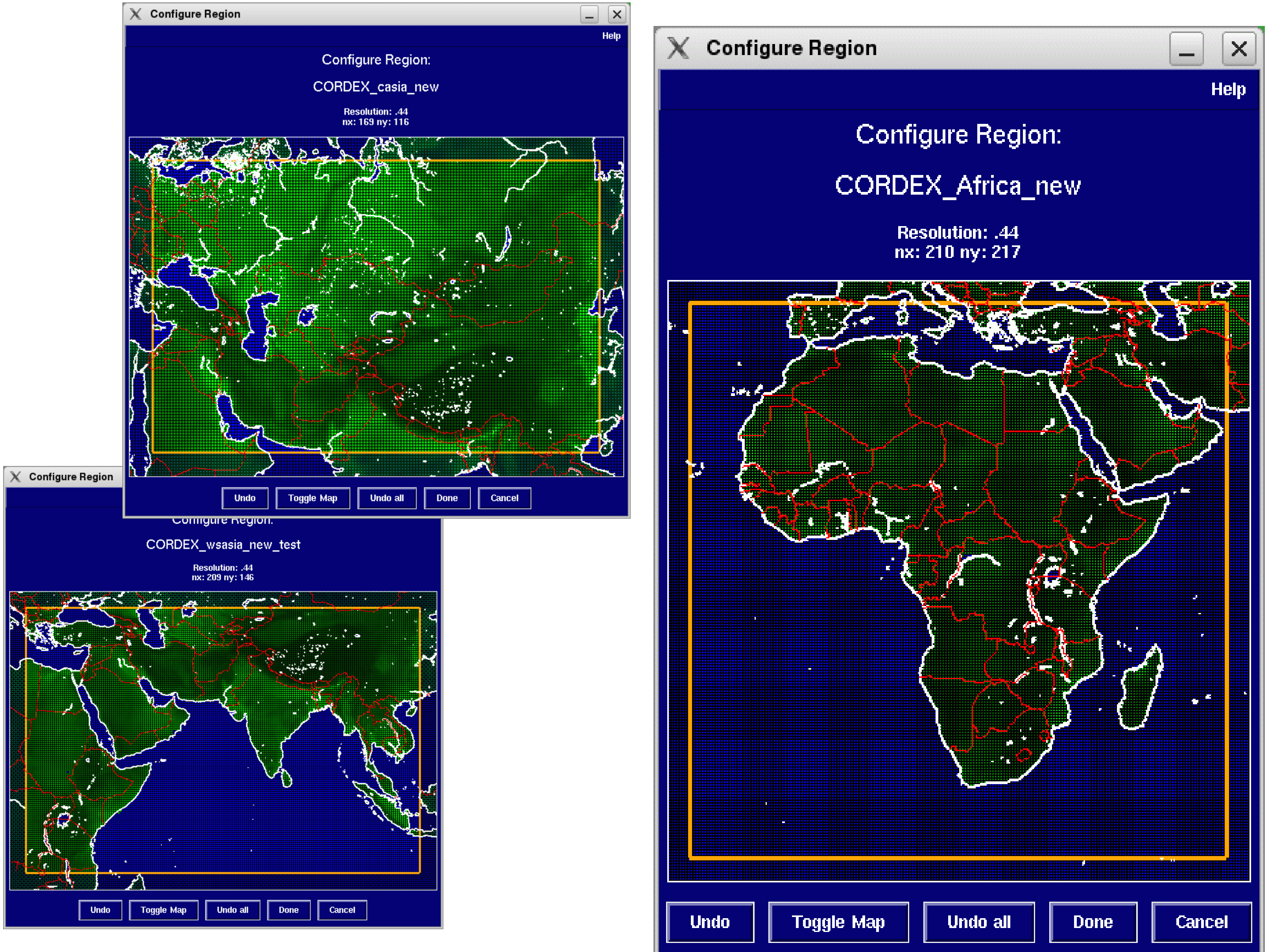
- **Fig. 4.7 Example of improvement in simulation of climatological precipitation as resolution of the climate model is increased from that of the general circulation model (T42, approximately 250km resolution) to regional model at 80km resolution (1st nest) to regional model at 20km resolution (2nd nest). (from Liqiang Sun, IRI).**



Increase annual max number consecutive dry days (<1 mm) Kamaguchi SOLA 2006



CORDEX – an integrated downscaling effort based on new modelling results



Message 10:

A huge – and often wasted effort -has been made on downscaling. Within the next few years there will be a major step forward in the quality and consistency of regional modelling

Some pragmatic advice about climate modelling

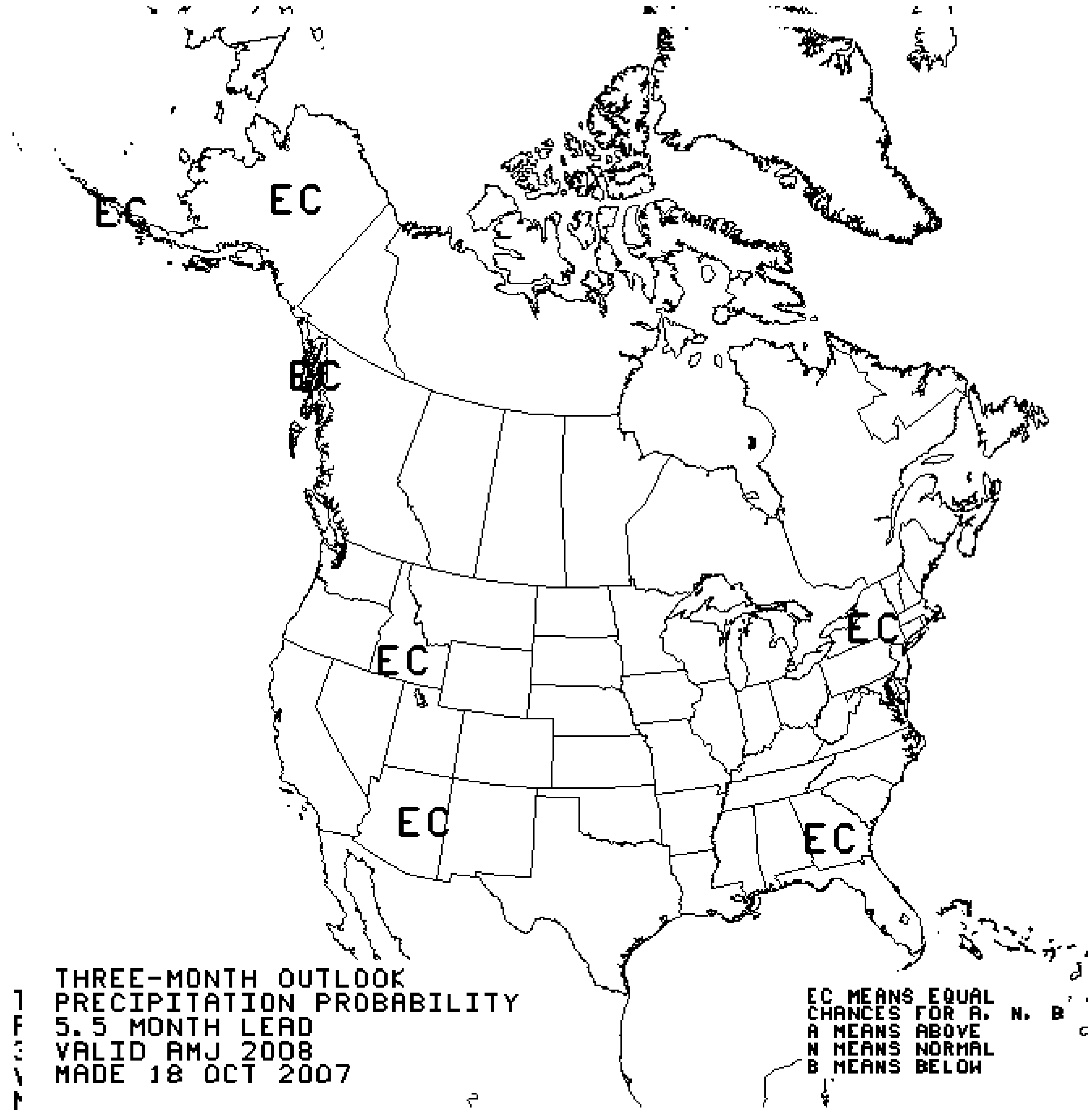
- A higher resolution model is not necessarily more skillful
- Don't pick winners - good predictions of current climate by a GCM does not guarantee good predictions of future climates
- Models may seem "right" for the wrong reason.
- Beware of model "loyalty" ... all models are wrong! But some are more useful than others – in some situations.
- Use a suite of projections, not just one
- Downscaling cannot improve poor GCM inputs
- Consider the relative merits of dynamic versus statistical downscaling
- Use common sense, e.g. is the spatial cohesion of the projected change sensible?
- Projecting the direction of change is more robust than its absolute magnitude
- What information would cause you to change your plans?
- Out to 2050, the choice of the emissions scenario is not hugely important (i.e. A2 or B2 SRES etc produce similar results).

- Based on original by Bruce Hewitson – Uni Capetown

Seasonal forecasting NOAA

- 1, 3 & 5.5 month precipitation forecast
- But essentially little useable information beyond 1 month

Which "Early Warning Systems" are feasible?



The Catarina Mystery

28th March 2004

Waters “too cold”

Moved East to West – wrong direction

No similar landfall on record

BUT

Similarities to phenomena on east coast of Australia

But there will be surprises

Message 11:
But there will always be things that remain unknowable

And we will continue to live with surprise

Robust decision making approaches are needed

Important similarities

and differences

between DRR & CCA

Issues to be covered

Is the changing climate affecting disaster risk and risk management?

What can we learn from climate science and climate modelling?

What is to come?

What do these projections mean for DRM?

Limits to what we can know and the downscaling issue

Important similarities and differences between DRR & CCA

Relationship between Development, DRR and CCA

- Failure to manage disaster risk is clearly a threat to development goals
- Climate change expands and changes the challenges of DRR
  - Climate related disasters become worse in most parts of the world
  - Chronic "bad weather" reduces coping capacity
- Climate change and its impacts changes the "moral grounds" of the development process

Relationship between development and CCA

Figure 7. A Continuum of Adaptation Activities: From Development to Climate Change

1 Addressing Drivers of Vulnerability

UGANDA: Providing women with crossbred goats and instruction in graze-free feeding *(Karamoja Agropastoral Development Programme)*

BANGLADESH: Diversification of livelihood strategies in areas vulnerable to flooding (**South***South*North)

CUBA: Vaccination program to eradicate diseases in low-income areas *(Cuban Ministry of Health)*

2 Building Response Capacity

BRAZIL: Paricipatory reforestation in Rio de Janeiro's hillside favelas to combat flood-induced landslides *(City of Rio de Janeiro)*

MONGOLIA: Reinstating pastoral networks to foster appropriate rangeland management practices in arid regions *(National University of Mongolia)*

TANZANIA: Reviving traditional enclosures to encourage vegetation regeneration and reduce land degradation *(Ministry of Natural Resources and Tourism, Tanzania)*

3 Managing Climate Risk

TANZANIA: Monitoring salinization of drinking water and drilling new wells to replace those that are no longer usable (**South***South*North)

MALI: Teaching farmers to collect climate data and integrate it into their planting decisions *(Government of Mali / Swiss Agency for Development and Cooperation)*

BANGLADESH: Using nationally standardized risk assessment procedures to develop a community adaptation plan of action *(local government)*

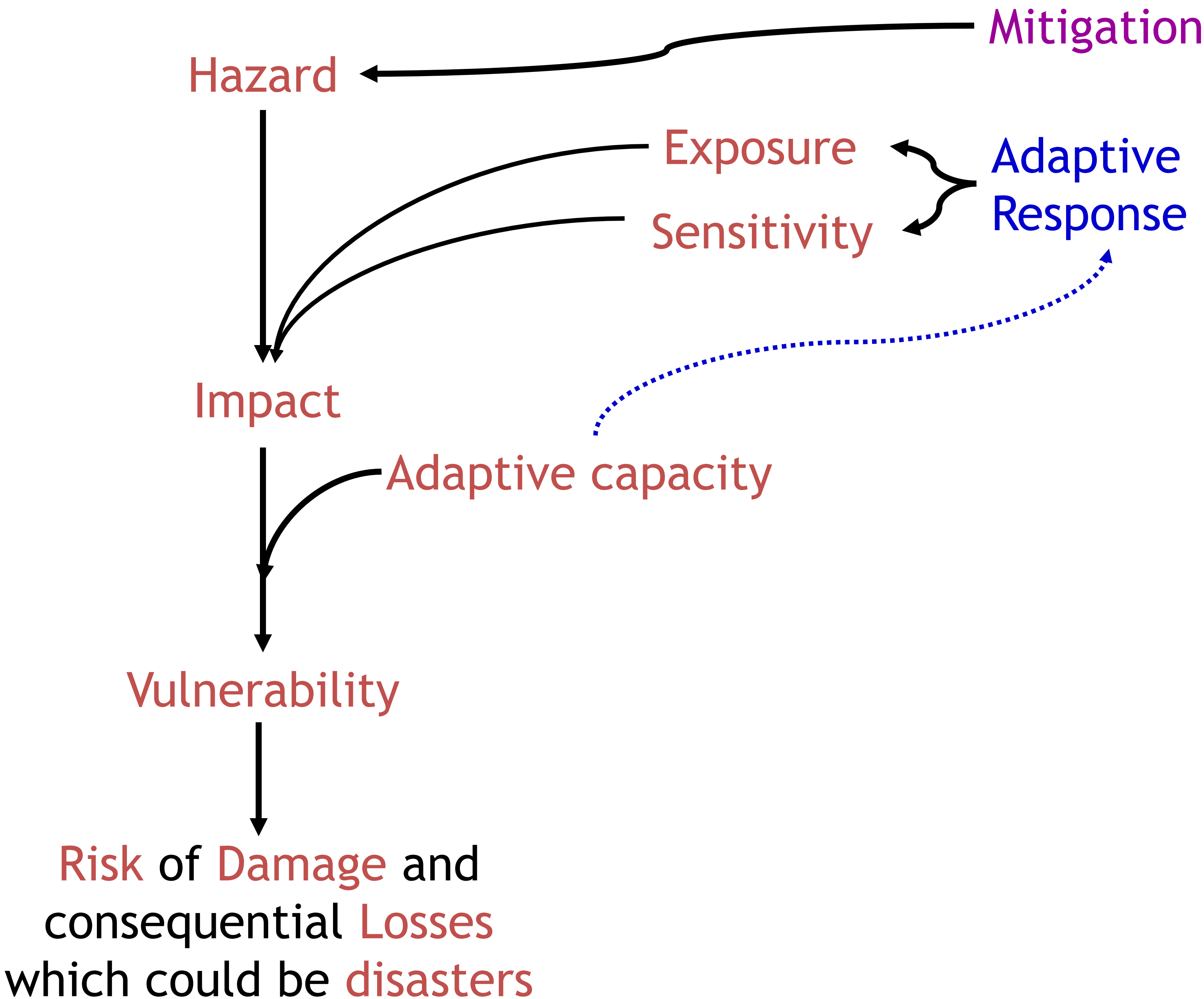
4 Confronting Climate Change

INDONESIA: Managing coral reefs in response to widespread coral bleaching *(WWF)*

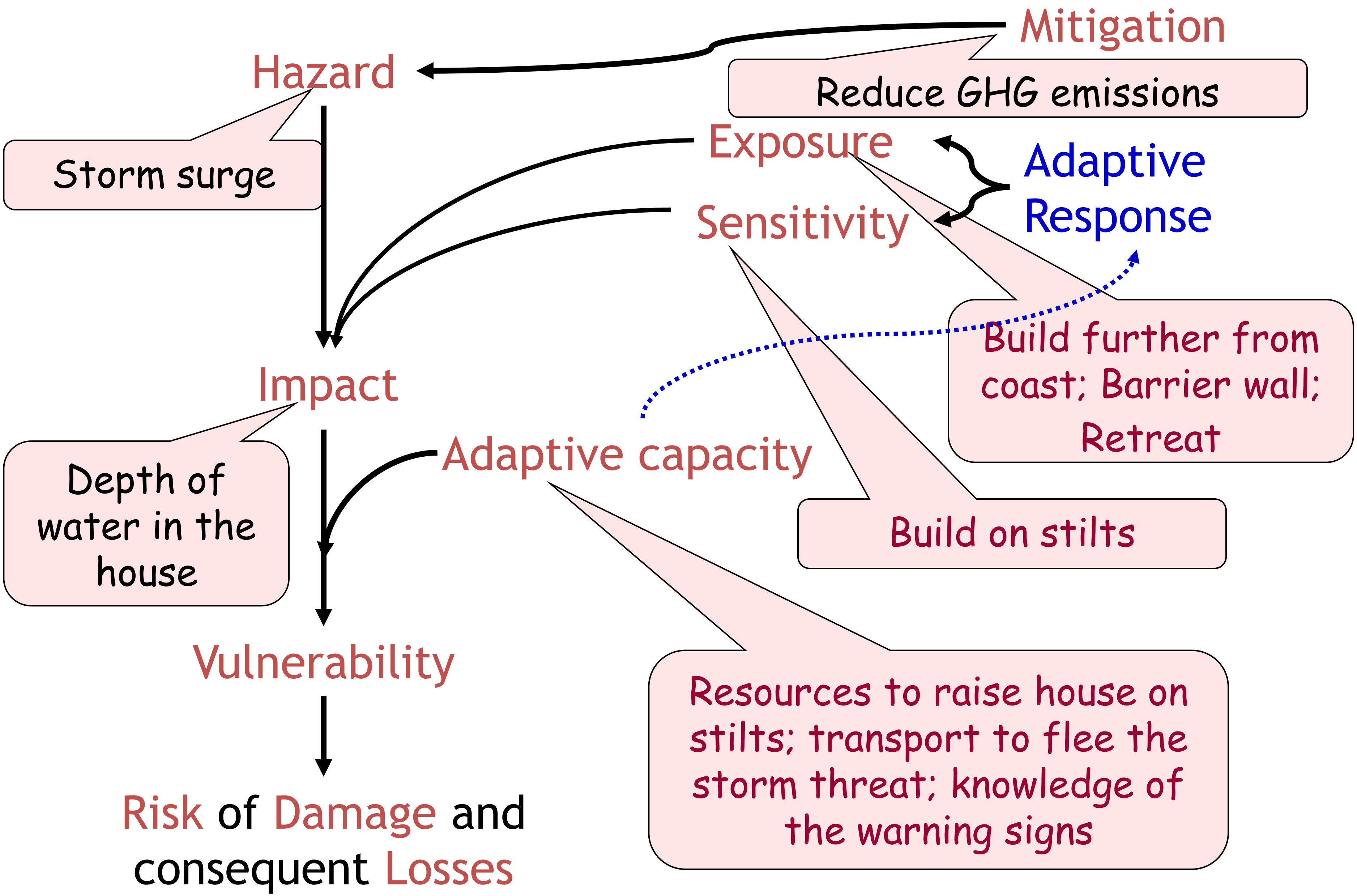
NEPAL: Reducing the risk of glacial lake outburst floods from Tsho Rolpa Lake *(Government of Nepal)*

VULNERABILITY FOCUS ← → IMPACTS FOCUS

World Resources Institute 2009


Mitigation
Hazard
Exposure
Adaptive
Response
Sensitivity
Impact
Adaptive capacity
Vulnerability
Risk of Damage and
consequential Losses
which could be disasters

Vulnerability of a coastal house to climate change



Commonalities and Differences … A starter

DRR	CCA
CC is just one of many factors that are changing	Similar
There will be more climatically extreme events	Similar
These will affect coping capacity of communities and nations	Chronic low level events will damage livelihoods
Less affected	Responsibilities and blame will confuse progress

Contact

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